



International corporate bond returns: Uncovering predictability using machine learning

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ABSTRACT

We examine the cross-sectional predictability of corporate bond returns using a novel international dataset and a set of machine learning techniques. We find strong predictability in both U.S. and non-U.S. markets, with differing predictive factors. Bonds in developed markets show greater integration with the U.S. market and stronger ties to equity markets. Predictive performance of machine learning models varies over time and is greater before the onset of the COVID-19 pandemic and during periods of deteriorating business conditions, reduced market liquidity, elevated investor sentiment, and heightened risk aversion. The results offer insights into bond pricing and global diversification opportunities.

1. Introduction

This paper provides the first study to employ machine learning (ML) techniques to examine the cross-sectional predictability of corporate bond returns in international markets. Prior studies have exclusively focused on the U.S. market (Fama and French, 1993; Huang and Shi, 2021; Bali et al., 2022; Feng et al., 2025). However, over the past few decades, international corporate bond markets have increasingly been influenced by non-U.S. markets, particularly emerging ones. Fig. 1 shows that the outstanding value of emerging markets surged from under \$1 trillion in 2005 to over \$15 trillion in 2022, accounting for 3 % and 26 % of the international corporate bond market's outstanding values, respectively. Although there is growing importance of non-U.S. corporate bond markets, research in this area is limited, largely due to the scarcity of data on international corporate bonds.¹ Non-U.S. corporate bond markets, especially emerging markets, can be fundamentally different from the U.S. market due to incomplete information, weaker investor protection, and lower financial stability. Understanding these differences and identifying the factors that drive bond pricing across

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¹ It is only in recent years that researchers have begun to investigate bond securities outside the U.S. (Bekaert and De Santis, 2021; Li et al., 2023).

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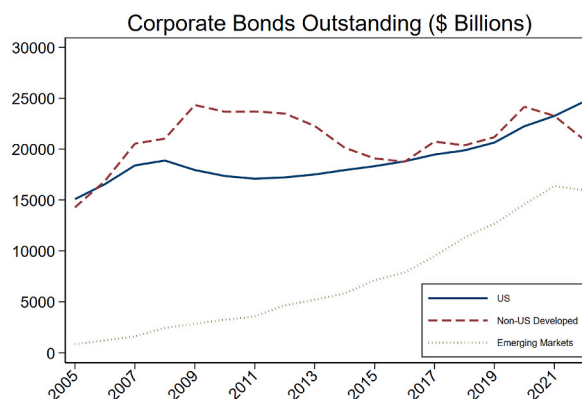


Fig. 1. International corporate bond market development, 2005–2022. This figure illustrates the growth of the international corporate bond market in terms of outstanding bond values, measured in billions of U.S. dollars, from 2005 to 2022. The markets are categorized into three groups: the U.S. market, non-U.S. developed markets, and emerging markets. The data are obtained from the Bank of International Settlements (BIS).

markets is of critical importance to researchers and policymakers. This issue is also highly relevant to practitioners, as large asset management firms have increasingly introduced investment vehicles with exposure to international corporate bonds.

Given these developments, there is a pressing need for modeling approaches that can account for the structural complexity of global corporate bond markets. These markets are characterized by substantial heterogeneity in issuer characteristics, complex contractual features, and a high-dimensional set of predictive signals. Such complexities give rise to nonlinearities and informational frictions that traditional linear models often fail to capture. ML techniques are particularly well-suited to this environment, as they can accommodate high-dimensional data, model intricate interactions, and identify inefficiencies arising from delayed price discovery and credit-specific risks.

We use data from IHS Markit, which provides high-quality pricing information for a large set of international corporate bonds. We match this data with Bloomberg and Datastream to construct a unique dataset of approximately 24,450 USD-denominated corporate bonds issued by companies in 86 economies between 2012 and 2021. This dataset allows us to calculate bond returns and characteristics, as well as the stock prices and firm-level characteristics of the corresponding companies.² Bond characteristics include (i) fundamental characteristics (e.g., age, illiquidity, and credit rating); (ii) return-based characteristics (e.g., short-term reversal, momentum, and volatility); and (iii) exposure to some common asset pricing factors (e.g., *TERM*, *DEF*, *BMKT*, *CRF*, *LRF*, and *DRF*).

We apply cutting-edge ML techniques to our dataset to study the cross-sectional predictability of corporate bond returns. Importantly, we aim to distinguish the return predictability outside the U.S. from that within the U.S. We start by considering 21 bond characteristics as potential return predictors and categorize the economies in our sample into three markets: U.S., non-U.S. developed, and emerging markets. We then follow Gu et al. (2020) to use 10 different ML models, including penalty estimation, dimension reduction approaches, regression tree models, and neural network-based methods, to predict monthly corporate bond returns.³ We further examine the predictive performance of these ML methods, comparing linear models to non-linear models to assess the advantages of accounting for potential nonlinearities. This approach enables us to determine whether incorporating non-linear relationships improves return predictability across different corporate bond markets.

We find that all 10 ML models show evidence of predictabilities with reasonable, positive out-of-sample *R*-squared values both within and outside the U.S. market. However, the predictive performance of these models is different cross markets, potentially due to distinctive market structures. For example, nonlinear models generally outperform linear models in the U.S. market, generating higher *R*-squared values, while linear models show better performance in non-U.S. markets. Moreover, the improvement in predictive power is more pronounced for investment-grade bonds in developed markets, while the predictive performance is stronger for non-investment-grade bonds in emerging markets.

The portfolio analysis shows that all 10 ML models, when applied to the U.S. market, generate both economically and statistically significant excess returns, ranging from 9.8 % to 14.0 % per year. An annualized Sharpe ratio varies from 1.09 to 1.56, based on a long-short portfolio strategy that uses predicted bond returns. For corporate bonds outside the U.S., the ML models instead generate an annualized Sharpe ratio ranging from 0.48 to 1.60. The investment gains remain robust after accounting for transaction costs, with moderate monthly turnover ratios for the long-short portfolio strategy staying below 70 % across all three markets.

² Using the same filtering criteria as in this study, we find that while non-U.S. developed markets yield 156,300 observations in local currencies (compared to 97,758 in USD-denominated bonds), emerging markets produce only 2950 observations in local currencies, compared to 81,282 in USD-denominated bonds. This sharp reduction in coverage for emerging markets is primarily due to the limited availability of credit rating information for local-currency bonds in Datastream. As a result, including local-currency bonds would undermine our ability to conduct meaningful and consistent cross-market comparisons. Therefore, by focusing on USD-denominated bonds—a subset of the broader international corporate bond market—our analysis may be more reflective of bonds issued by internationally active firms or those accessing global capital markets.

³ Kelly and Xiu (2023) provide a review of the literature on financial machine learning and AI.

In addition, we find that bond returns are predicted by distinct characteristics and factors in different markets. Momentum, downside risk, and illiquidity are the most influential predictors in the U.S. market, non-U.S. developed markets, and emerging markets, respectively. However, these three markets also share some similarities in terms of predictive characteristics. For example, downside risk, momentum, and exposures to certain bond risk factors (especially default beta) are among the 10 most important predictive characteristics in both the U.S. and non-U.S. markets. The similar reliance on characteristics and factors in bond pricing suggests that the characteristic–return relationship could be somewhat consistent across these three markets. This result aligns with [Hellum et al. \(2024\)](#), who find transfer-learning that predictive models for global equity markets are largely similar. To further assess model similarity across markets, we train our ML models using U.S. data to predict returns in the non-U.S. developed markets and emerging markets. The results show that in emerging markets, the U.S.-trained models substantially underperform compared to models trained by local data, with lower *R*-squared values for all the ML models. However, in non-U.S. developed markets, three out of the ten U.S.-trained models have higher *R*-squared values than those trained by local data. These results indicate that, compared to other developed markets, the characteristic–return relation in emerging markets differs more significantly from that in the U.S., likely due to the lower liquidity of bond trading in these markets.

To further examine the cross-market integration of corporate bonds, we include exposure to the five most common U.S. bond factors (i.e., *BMKT*, *DEF*, *CRF*, *LRF*, and *DRF*) as additional characteristics for non-U.S. corporate bonds. We aim to determine whether including U.S. exposure can improve the predictive performance of ML models for bond returns outside the U.S. This method is motivated by the canonical international finance literature that an asset's returns are more tied to global factors than local factors if financial markets are more integrated ([Chari and Henry, 2004](#); [Pukthuanthong and Roll, 2009](#)). Our results show that most models generate higher *R*-squared after adding U.S. exposures in both non-U.S. developed markets and emerging markets, indicating that international corporate bond markets are integrated.

To investigate whether explicit and implicit barriers lead to cross-economy differences in market integration, we calculate, for each economy, the increase of *R*-squared after adding U.S. exposures and use it as a measure of economy-level integration between an economy's corporate bond market and the U.S. corporate bond market. A larger increase in *R*-squared indicates a higher level of integration. We find that, on average, non-U.S. developed markets are more integrated with the U.S. corporate bond market compared to emerging markets. However, there are still significant differences in integration across different non-U.S. developed markets and across different emerging markets. Additionally, the correlation between cross-market integration in bond markets and equity markets is relatively low.

We also examine bond-stock integration across different markets by investigating whether stock characteristics can help improve the cross-sectional return predictability of corporate bonds. Following [Bali et al. \(2022\)](#), we introduce 24 stock characteristics into our 10 models, in addition to the existing 21 bond predictors. We find that in the U.S. market, all 10 models experience a drop of *R*-squared after adding stock characteristics, indicating that corporate bonds and stocks are not highly integrated in the U.S. By contrast, in non-U.S. markets, the predictive performance increases for two of the 10 models when stock characteristics are added to the models. This result suggests a certain degree of bond-stock integration outside the U.S. market. Our economy-level analyses further show that, on average, developed markets reveal higher levels of bond-stock integration than emerging markets.

Lastly, we investigate the time-varying nature of predictive performance. We find that the performance of our 10 ML models declines substantially across all three markets following the onset of COVID-19, suggesting a structural shift in the informativeness of traditional bond characteristics. Moreover, bond return predictability is notably higher during periods of deteriorating business conditions, reduced market liquidity, elevated investor sentiment, and heightened risk aversion. Among the four macroeconomic variables, investor sentiment emerges as the most influential factor in the two developed markets, whereas market liquidity plays the most prominent role in emerging markets. These findings are consistent with prior evidence from the U.S. equity, Treasury bond, and corporate bond markets ([Dangl and Halling, 2012](#); [Bianchi et al., 2021](#); [Avramov et al., 2023](#); [Feng et al., 2025](#)), and they offer novel insights from a global corporate bond perspective.

This paper contributes to three strands of literature. First, we add to the growing literature on the predictability of corporate bond returns. Previous papers examine various bond characteristics, including default and term betas ([Fama and French, 1993](#); [Gebhardt et al., 2005](#)), momentum ([Jostova et al., 2013](#)), liquidity ([Lin et al., 2011](#); [Bongaerts et al., 2017](#)), volatility ([Chung et al., 2019](#); [Bao et al., 2015](#)), bond book-to-market ratio ([Bartram et al., 2025](#)), long-run consumption growth ([Elkamhi et al., 2024](#)), and long-term reversal ([Bali et al., 2021](#)). However, all these studies primarily focus on the U.S. market, with limited research on international corporate bond markets. Examining international markets, [Valenzuela \(2016\)](#) finds that market liquidity contributes to credit spreads, and [Lee et al. \(2022\)](#) show that forward rates contain information about corporate bond returns. [Bekaert and De Santis \(2021\)](#) identify ratings, maturity, and bond market beta as key characteristics explaining corporate bond returns in major developed markets. In this paper, we use a novel dataset of international corporate bonds and focus on out-of-sample prediction, rather than in-sample tests. [Goyal and Welch \(2008\)](#) argue that out-of-sample tests provide the most rigorous and relevant evidence of return predictability. Furthermore, we employ ML techniques to address the high dimensionality introduced by the large number of predictive variables.

Second, this paper extends and complements the literature that predicts asset returns using ML. While most existing studies focus on the stock market,⁴ a growing number have begun to examine other asset classes. For instance, [Bianchi et al. \(2021\)](#) find that non-linear ML models, such as regression trees and neural networks, outperform linear techniques in predicting U.S. Treasury bond returns. Several studies also apply ML methods to predict corporate bond returns in the U.S. market ([Lin et al., 2018](#); [Bali et al., 2022](#); [Feng](#)

⁴ These papers include, but are not limited to, [Feng et al. \(2020\)](#), [Freyberger et al. \(2020\)](#), [Gu et al. \(2020\)](#), [Kozak et al. \(2020\)](#), [Giglio et al. \(2021\)](#), [Leippold et al. \(2022\)](#), [Dong et al. \(2022\)](#), and [He et al. \(2023\)](#).

et al., 2025). Unlike these studies, we use ML models to study corporate bond return predictability in international markets and find major economic differences in return predictors among U.S., non-U.S. developed, and emerging markets.

Third, our paper expands the research on the integration of international financial markets. There has been a longstanding debate regarding whether stocks are priced locally or globally. Most studies find that cross-sectional returns are better explained by local factors (Griffin, 2002; Bekaert et al., 2009; Hou et al., 2011; Chaieb et al., 2021b; Hollstein, 2022). To the best of our knowledge, only Bekaert and De Santis (2021) examine the integration of international corporate bond markets. Our paper differs from theirs in two ways. First, they focus solely on integration across major developed economies, whereas we include a large number of emerging markets. Second, we incorporate a vast number of bond characteristics that contain important predictive information. To manage this large set of predictors, our use of state-of-the-art ML models makes it possible to gain new economic insights.

The remainder of the paper is structured as follows. In Section 2, we provide a detailed description of our dataset and outline the research methodology. In Section 3, we discuss the main findings. In Section 4, we report additional analyses and robustness tests. We conclude in Section 5.

2. Data and methodology

2.1. Data, corporate bond returns, and predictors

We obtain daily data on corporate bond pricing from IHS Markit, covering the period from February 2010 to December 2021. While transaction-level data are available in the U.S. through databases such as TRACE, transaction prices for corporate bonds outside the U.S. are typically not available. Instead, the bond pricing provided by the Markit dataset is generally based on dealers' quotes or asset pricing models, such as matrix pricing. Despite this, these prices are widely regarded as a fundamental measure of bond valuation (e.g., Friewald et al., 2012; Schestag et al., 2016). The Markit dataset provides a wide range of pricing information, including bid, mid, and ask prices. Additionally, it includes multiple types of yields and various bond features, such as coupon rates, country of risk, issuance amount, and bond grades (investment-grade or high-yield).⁵ Furthermore, it includes issuer information and complex bond metrics, such as duration and option-adjusted spreads, which are derived through sophisticated calculations. This comprehensive dataset thus enables a detailed analysis of corporate bond pricing and characteristics across different markets and periods.

To obtain the equity International Securities Identification Number (ISIN), we link the Markit data to Bloomberg using each bond's ISIN. Credit ratings are sourced from Datastream. To facilitate a fair comparison of corporate bonds across different regions, we categorize markets into three groups: the U.S., non-U.S. developed, and emerging markets.

We apply the following criteria to filter the data: (1) retaining only bonds denominated in USD to mitigate the impact of volatile exchange rates; (2) excluding duplicated bonds issued under Rule 144A if their Regulation S counterparts are already included in the sample; (3) removing perpetual, inflation-adjusted, or defaulted bonds; (4) discarding bonds with a maturity of less than six months; (5) eliminating bonds with a midprice below \$5 or above \$1,000; (6) excluding bonds without credit ratings; (7) removing bonds without an equity ISIN; and (8) retaining only bonds with a fixed or zero coupon.

Table 1 presents the summary statistics of our final sample, which consists of 945,562 monthly observations of 24,450 USD-denominated corporate bonds issued by 2,215 companies. The distribution of these observations across different markets reveals that the U.S. market dominates the sample, accounting for 81 % of the total observations; non-U.S. developed markets contribute 10 %, and emerging markets account for the remaining 9 %. These numbers are consistent when examining the distribution of the number of bonds: 78 % of the bonds originate from the U.S. market, 12 % from non-U.S. developed markets, and 10 % from emerging markets.

Interestingly, while the U.S. market constitutes a large share of the bonds and observations, only 58 % of the 2,215 companies in the sample are based in the U.S. This suggests that U.S. firms are more likely to issue a greater number of bonds compared to firms from non-U.S. markets. This observation is particularly notable given the various sources of corporate bond data, yet the percentage of U.S. firms in our study (58 %) aligns closely with the 57 % reported in Wang (2023). This consistency further validates the representativeness and reliability of our sample in capturing the dynamics of the USD-denominated corporate bond market within the broader global context.

The monthly bond return is calculated as:

$$r_{i,t} = \frac{P_{i,t} + AI_{i,t} + C_{i,t}}{P_{i,t-1} + AI_{i,t-1}} - 1, \quad (1)$$

where $P_{i,t}$ denotes midprice, $AI_{i,t}$ represents the accrued interest, and $C_{i,t}$ is the coupon payment for bond i in month t . We denote $R_{i,t}$ as bond i 's return in excess of the U.S. risk-free rate in month t . Returns are calculated using the last available price observation of month t ($P_{i,t}$) and the last available price observation of month $t - 1$ ($P_{i,t-1}$), allowing for non-trading days.

We incorporate a total of 21 bond characteristics and 24 stock characteristics to predict corporate bond returns.⁶ Appendix A

⁵ Country of risk takes into account the issuing company's country of domicile, country of listing, country of largest revenue, and reporting currency (most important to least important). See the documents provided by Bloomberg L.P. for more details.

⁶ Out of our 21 bond predictors, 11—such as volatility and bond market beta—are constructed using observations from the past 36 months. We require at least 24 monthly return observations within each 36-month rolling window. Since our data begins in February 2010 and one month is lost for return calculation, our final sample spans roughly 10 years, from March 2012 to December 2021.

Table 1

Sample distribution, This table reports the sample distribution across three market groups: U.S., non-U.S. developed, and emerging markets. *Number of obs.* indicates the number of monthly observations. *Number of bonds* refers to the number of unique bond ISINs. *Number of companies* indicates the number of unique equity ISINs corresponding. Percentages in parentheses indicate each group's share of the full sample.

	Number of obs.	Number of bonds	Number of companies
U.S.	766,522 (81 %)	19,103 (78 %)	1,285 (58 %)
Non-U.S. developed	97,758 (10 %)	2,894 (12 %)	302 (14 %)
Emerging	81,282 (9 %)	2,453 (10 %)	628 (28 %)
Total	945,562	24,450	2,215

provides a detailed description of these 45 variables, including the data sources used. We obtain stock pricing and trading information, as well as accounting variables, from Datastream. To ensure consistency and comparability across the three markets, we normalize all characteristics to have a zero mean and a unit standard deviation monthly. For any missing data, we adopt the methods outlined by [Bali et al. \(2022\)](#) and [Gu et al. \(2020\)](#), replacing missing values with the cross-sectional median for each month. This approach allows us to maintain the integrity of the data while addressing gaps in the dataset.

[Table 2](#) presents the time-series average of the cross-sectional distribution of bond returns and several key characteristics, including credit rating, time-to-maturity, age, duration, size, illiquidity, and downside risk. On average, the monthly observations are 6,182 for the U.S. market, 788 for non-U.S. developed markets, and 656 for emerging markets. The average monthly bond return in the U.S. market is 0.28 %, while non-U.S. developed markets exhibit a slightly lower average return of 0.24 %. The highest average monthly return is 0.30 % for emerging markets, indicating greater compensation for the higher risks associated with these markets. The magnitude of these returns for our international corporate bond database aligns well with other global corporate bond data sources. For example, [Bekaert and De Santis \(2021\)](#) report an annualized mean return of 3.58 % for BBB-rated bonds.

The higher returns in emerging markets reflect the higher risks associated with these markets, as evidenced by several risk measures. For example, credit risk is highest in emerging markets, with an average score of 9.78, compared to 8.44 in the U.S. market and 7.13 in non-U.S. developed markets. We report credit ratings using conventional linear numerical scores, where a higher score indicates greater credit risk (e.g., AAA = 1, AA+ = 2, ..., C = 21). Emerging markets also exhibit the highest level of illiquidity among the three regions, with an average illiquidity score of 0.72, compared to 0.59 and 0.47 in the U.S. and non-U.S. developed markets, respectively. Emerging markets also show the highest downside risk, as measured by the 5 % value-at-risk (VaR5), which represents the second-lowest monthly return over the past 36 trading months. Moreover, bond returns in emerging markets are more prone to extreme by observing the 1st and 99th percentile returns, indicating a greater likelihood of experiencing significant losses and gains. This underscores the volatile nature of returns in these markets compared to developed markets.

2.2. Methodology

In its most general form, we express corporate bond excess return as the sum of expected returns and the error term with zero mean:

$$R_{i,t+1} = E_t(R_{i,t+1}) + e_{i,t+1}, \quad (2)$$

where bonds are indexed as $i = 1, \dots, N$, and time periods, specifically months, are denoted by $t = 1, \dots, T$. $E_t(R_{i,t+1})$ refers to the expected excess return for bond i in month $t + 1$, conditional on the information available in month t . In the context of corporate bond returns, the data structure is notably imbalanced, with many bonds having sparse or incomplete observations across different time periods. This imbalance poses significant challenges for traditional modeling approaches. To address these challenges, we follow the methodology proposed by [Gu et al. \(2020\)](#), utilizing a panel data approach to predict bond returns. This approach assumes a homogeneous conditional expectation functional form, meaning that the relationship between bond returns and their predictors is consistent across all observations, despite the imbalances in the data.

Our primary objective is to capture the dynamics of individual corporate bond excess returns by using a comprehensive set of predictors. These predictors, detailed in [Appendix A](#), include various characteristics of the bonds, such as maturity, age, duration, size, illiquidity, and downside risk. The central task is to estimate a functional form $g(z_{i,t})$ that connects the expected future bond returns $E_t(R_{i,t+1})$ to characteristics, $z_{i,t}$ of bond i :

$$E_t(R_{i,t+1}) = g(z_{i,t}). \quad (3)$$

Building on the work of [Gu et al. \(2020\)](#) and [Bali et al. \(2022\)](#), our study focuses on four distinct categories of ML models: penalized linear models (LASSO, RIDGE, and ENet), dimension reduction techniques [Principal Component Regression (PCR) and Partial Least Squares (PLS)], regression trees [Random Forest (RF) and Gradient Boosted Regression Trees (GBRT)], and neural networks (NN1, NN2, and NN3). Each of these models represents a different approach to handling the complexities inherent in predicting corporate bond returns, with varying degrees of sophistication in addressing linearity and nonlinearity in the data.⁷

To further investigate the potential advantages of incorporating nonlinearity in forecasting corporate bond returns, we classify the

⁷ The detailed description of these methods and the choice of hyperparameter can be found in [Gu et al. \(2020\)](#).

Table 2

Descriptive statistics, This table reports the number of bond-month observations, as well as the cross-sectional mean, median, standard deviation, and percentiles of monthly corporate bond returns, credit ratings, time to maturity, age, duration, size, illiquidity, and downside risk in the U.S. market (Panel A), non-U.S. developed markets (Panel B), and emerging markets (Panel C). Credit ratings are expressed as conventional linear numerical scores, where a higher numerical score indicates higher credit risk (e.g., AAA = 1, AA+ = 2, ..., C = 21). Size is measured as the natural logarithm of the amount issued. Following Amihud and Mendelson (1986), illiquidity is defined by the relative bid-ask spread, divided by the average of bid and ask prices. Downside risk is proxied by the 5 % value-at-risk (VaR5), representing the second-lowest monthly return over the past 36 trading months.

	N	Mean	Media	SD	1st	p5	p25	p75	p95	99th
Panel A: U.S.										
Bond return (% monthly)	766,522	0.28	0.19	2.38	-5.37	-2.34	-0.56	1.05	3.07	6.33
Rating (AAA = 1, AA+ = 2, etc.)	766,522	8.44	8.04	3.09	1.82	4.68	6.34	9.80	14.64	17.50
Time to maturity (years)	766,377	9.56	6.21	9.07	0.62	1.06	3.23	14.55	25.81	31.06
Age (years)	766,298	8.00	6.28	5.75	2.08	2.36	3.86	9.57	20.96	25.31
Duration	727,672	5.48	4.00	4.58	0.21	0.50	1.72	8.67	14.51	16.55
Size	762,288	5.26	5.84	1.85	0.60	1.41	4.46	6.48	7.45	8.01
Illiquidity	760,008	0.59	0.49	0.71	0.03	0.06	0.22	0.83	1.39	2.03
Downside risk (5 % VaR)	766,377	3.22	2.59	2.75	0.34	0.61	1.47	4.16	7.97	13.29
Panel B: Non-U.S. developed										
Bond return (% monthly)	97,758	0.24	0.16	1.95	-4.37	-1.96	-0.41	0.85	2.65	5.44
Rating (AAA = 1, AA+ = 2, etc.)	97,758	7.13	6.92	3.03	1.04	3.17	5.00	8.82	12.91	16.61
Time to maturity (years)	97,748	7.26	4.74	7.76	0.57	0.80	2.20	8.37	24.37	27.48
Age (years)	97,711	6.77	5.16	4.84	2.06	2.24	3.30	8.53	17.74	23.11
Duration	94,962	4.58	3.40	4.04	0.30	0.49	1.47	6.31	13.54	15.77
Size	97,248	5.80	6.27	1.71	0.32	1.90	5.25	6.87	7.53	7.82
Illiquidity	96,872	0.47	0.35	0.49	0.03	0.05	0.16	0.67	1.23	1.92
Downside risk (5 % VaR)	97,748	2.69	2.06	2.37	0.27	0.52	1.13	3.48	7.10	11.36
Panel C: Emerging										
Bond return (% monthly)	81,282	0.30	0.26	2.80	-8.31	-2.63	-0.43	1.03	3.35	8.48
Rating (AAA = 1, AA+ = 2, etc.)	81,282	9.78	9.58	3.39	3.60	4.80	7.49	11.92	15.83	18.25
Time to maturity (years)	81,259	6.00	3.76	7.19	0.60	0.85	2.07	6.63	23.03	27.37
Age (years)	81,259	5.24	4.11	3.59	2.05	2.19	2.94	6.36	12.61	20.51
Duration	80,334	3.79	2.63	3.48	0.27	0.48	1.36	5.09	12.17	14.95
Size	81,111	6.24	6.22	0.60	4.62	5.25	5.89	6.63	7.17	7.62
Illiquidity	80,372	0.72	0.58	0.84	0.05	0.10	0.36	0.81	1.66	3.58
Downside risk (5 % VaR)	81,259	3.53	2.55	3.53	0.30	0.61	1.43	4.30	9.91	19.11

10 ML models into two broad groups: linear models and nonlinear models. The linear models, which include LASSO, RIDGE, ENet, PCR, and PLS, are designed to capture linear relationships between the predictors and the corporate bond returns. In contrast, the nonlinear models, consisting of RF, GBRT, and the three neural networks (NN1-NN3), are equipped to handle more complex, nonlinear interactions in the data. This categorization follows the approach outlined by Bali et al. (2023), who emphasize the importance of distinguishing between linear and nonlinear modeling techniques when forecasting financial asset returns. To synthesize the predictions from each category, we average the predictions of the individual models within each group. This process results in a combined prediction for the linear models (L-En) and a separate combined prediction for the nonlinear models (N-En). The benefit of such a prediction combination in improving economic forecasts has been well documented in the literature (e.g., Bates and Granger, 1969; Rapach et al., 2010; Lin et al., 2018).

In our empirical analysis, we divide the entire sample period into three distinct phases to ensure robust model estimation, validation, and testing. The first phase, consisting of the initial three years, serves as the training period (T_1), during which the models are estimated using historical data. The second phase, spanning the subsequent two years, acts as the validation period (T_2), where we fine-tune the hyperparameters of the models to optimize their performance. Finally, the third phase, covering the next five years, is designated as the test period (T_3), during which we conduct out-of-sample analysis to evaluate the models' predictive accuracy and performance in real-world scenarios.

Due to computational constraints, we retrain our models on an annual basis rather than monthly. Each time the model is refitted, we expand the training sample by one additional year while maintaining the validation sample at a constant size. This approach strikes a balance between updating the model with new data and preserving a stable validation framework for consistent hyperparameter tuning.

2.3. Out-of-sample performance evaluation

Following the methodology of Gu et al. (2020) and Bali et al. (2022), we use the out-of-sample R -squared (R_{OOS}^2) to evaluate the predictive power of ML models for corporate bond returns:

$$R_{OOS}^2 = 1 - \frac{\sum_{(i,t) \in T_3} (R_{i,t} - \hat{R}_{i,t})^2}{\sum_{(i,t) \in T_3} (R_{i,t})^2}, \quad (4)$$

where $R_{i,t}$ represents the actual observed excess return for bond i in month t , and $\hat{R}_{i,t}$ denotes one-month-ahead predictions of bond i in month t . This R^2_{OOS} measures the proportional reduction in mean squared forecast error relative to a naïve forecast benchmark of zero, assuming that the one-month-ahead expected bond return equals the risk-free rate.

To further analyze the impact of different predictors on the model's forecasting accuracy, we assess their relative importance. We follow the methodology outlined by Gu et al. (2020), which involves calculating the reduction in the time-series average of the monthly R -squared ($\bar{R}^2_{OOS,t}$) when a specific predictor is set to zero, while keeping all other predictors unchanged. This approach isolates the contribution of each predictor to the overall model performance. The variable importance of each predictor is then defined as the relative reduction in $\bar{R}^2_{OOS,t}$. Under this method, the characteristic that leads to the smallest reduction in $\bar{R}^2_{OOS,t}$ is assigned a variable importance of 0, while the characteristic that leads to the largest reduction is assigned a variable importance of 1.

3. Empirical results

3.1. Predictability of bond characteristics

We first examine the predictive performance of ML models that incorporate 21 bond characteristics. The primary goal of this study is to assess how well these models perform in both U.S. and non-U.S. markets.

Table 3 provides a comprehensive overview of the out-of-sample R -squared (R^2_{OOS}) results for all the models. Panel A presents R^2_{OOS} values for the U.S. market. In the first row, the R^2_{OOS} values are reported for all corporate bonds. The second and third rows provide results of an examination of the predictive performance of investment-grade and non-investment-grade bonds, respectively. For the full sample in the U.S. market, all ML models achieve positive R^2_{OOS} values, with results ranging from 3.53 % for GBRT to 4.61 % for NN3. Nonlinear (N-En) models outperform linear (L-En) models, with R^2_{OOS} of 4.5 % and 3.85 %, respectively. When comparing the R^2_{OOS} for all bonds, the results in the second row and third rows indicate that ML models exhibit superior predictive performance for investment-grade bonds, as evidenced by the higher R^2_{OOS} values. For example, the R^2_{OOS} for nonlinear (N-En) models in the U.S. investment-grade bond market is 3.56 %, while the R^2_{OOS} for linear (L-En) models in the non-investment-grade bond market is 2.25 %. This stronger predictive performance for investment-grade bonds in the U.S. market is consistent with the findings of Feng et al. (2025), highlighting the robustness of these models in such contexts.

Panels B and C of Table 3 provide results for non-U.S. markets, specifically developed markets and emerging markets, respectively. Consistent with the U.S. market findings, all ML models produce positive R^2_{OOS} for all bonds in non-U.S. developed markets and emerging markets. However, unlike in the U.S. market, linear (L-En) models outperform nonlinear (N-En) models in these two non-U.S. markets. In non-U.S. developed markets, as shown in Panel B, linear (L-En) models achieve an R^2_{OOS} of 4.38 %, while nonlinear (N-En) models lag behind with an R^2_{OOS} of 3.52 %. A similar trend is observed in emerging markets, where Panel C reveals that the R^2_{OOS} for linear (L-En) models is 5.17 %, surpassing the 4.81 % recorded for nonlinear (N-En) models. These results suggest that, in contrast to the U.S. market, linear relationships between predictors and corporate bond returns are more pronounced in non-U.S. markets, potentially due to different market dynamics and structural factors.

Furthermore, similar to the U.S. market, ML models still demonstrate higher R^2_{OOS} values for investment-grade bonds in non-U.S. developed markets, reinforcing the consistency of their performance across various market conditions. However, a notable deviation occurs in emerging markets, where non-investment-grade bonds exhibit better predictive performance. For example, in Panel C of

Table 3

Predictive performance of ML models. This table reports the out-of-sample predictive performance (R^2_{OOS}) of ML models in the U.S. market (Panel A), non-U.S. developed markets (Panel B), and emerging markets (Panel C). We use 21 bond characteristics to predict corporate bond returns across all three markets. For each market group, bonds are sorted into investment-grade bonds (with ratings of 10 or below, i.e., BBB- or better) and non-investment grade (with ratings of 11 or higher, i.e., BB + or worse). ML models include LASSO, ridge regression (RIDGE), elastic net (ENet), principal component analysis (PCA), partial least squares (PLS), random forest (RF), gradient boosted regression trees (GBRT), and neural networks with one to three layers (NN1–NN3). The L-En (N-En) model averages the forecasts of five linear (nonlinear) models, including LASSO, RIDGE, ENet, PCA, and PLS (RF, GBRT, NN1, NN2, and NN3).

	LASSO	RIDGE	ENet	PCA	PLS	RF	GBRT	NN1	NN2	NN3	L-En	N-En
Panel A: U.S.												
All bonds	3.62	3.67	3.67	4.16	3.82	3.76	3.53	4.16	4.39	4.61	3.85	4.50
Investment-grade	3.82	4.27	4.02	3.92	4.30	3.52	3.20	3.41	3.12	3.04	4.13	3.56
Non-investment-grade	1.63	1.07	1.68	3.06	2.04	1.61	2.08	2.80	1.19	1.88	2.04	2.25
Panel B: Non-U.S. developed												
All bonds	4.00	2.62	4.58	4.16	4.77	2.97	2.37	3.57	3.79	3.24	4.38	3.52
Investment-grade	4.53	4.31	4.55	4.65	5.03	5.04	3.84	3.65	3.63	3.51	4.82	4.33
Non-investment-grade	1.31	0.02	−0.11	3.12	0.52	0.98	1.96	0.88	1.82	0.72	1.56	1.66
Panel C: Emerging												
All bonds	4.99	5.14	4.97	5.02	5.28	4.75	3.01	4.78	4.91	4.67	5.17	4.81
Investment-grade	4.51	2.89	4.42	5.34	3.98	1.86	2.01	3.25	3.45	3.89	4.56	3.34
Non-investment-grade	4.57	4.86	4.66	4.23	4.68	4.58	2.28	4.48	4.36	4.15	4.75	4.48

Table 3, the R^2_{OOS} for nonlinear (N-En) models in the investment-grade bond segment of emerging markets is 3.34 %, which is lower than the 4.48 % observed for non-investment-grade bonds. These results suggest that, unlike in developed markets, ML models perform better in forecasting returns for non-investment-grade corporate bonds in emerging markets—possibly due to the greater complexity and information frictions associated with these markets.

3.2. Portfolio analysis

In addition to evaluating the predictive power of ML models for corporate bond returns across various markets, we extend our analysis to investigate the economic significance of these models through portfolio analysis. Specifically, at the end of each month, we categorize bonds within three different markets into deciles based on the forecasts of bond returns for the subsequent month. Following this, we construct both equal-weighted long-short portfolios consisting of corporate bonds, and value-weighted portfolios, where the weights are determined by the amount of bond issuance.

Table 4 provides a detailed summary of the equal-weighted monthly portfolio performance for all bonds, including both investment-grade and non-investment-grade categories. We report the average monthly returns for deciles ranging from decile 1 ("Low") to decile 10 ("High"), and we define the long-short portfolio strategy as "High-Low." In addition to this, we present the annualized Sharpe ratio for the long-short portfolios, which serves as a key measure of risk-adjusted performance.

Panel A of Table 4 provides results for an examination of the U.S. market, where a long-short trading strategy derived from ML model forecasts generates both economically and statistically significant performance. The average monthly returns (with corresponding annualized Sharpe ratios) for these strategies range from 0.82 % (1.09) to 1.17 % (1.56). Among the 10 ML models we analyze, the Principal Component Analysis (PCA) model and the Random Forest (RF) model exhibit the highest long-short trading strategy returns, at 1.17 % and 1.14 %, respectively.

Furthermore, when exploring the impact of incorporating nonlinearity into the models, our findings reveal that nonlinear (N-En) models outperform linear (L-En) models, achieving higher average monthly returns of 1.08 % and 1.14 %, respectively. In terms of the annualized Sharpe ratio for the long-short portfolios, the two neural network models, with one and two layers respectively, generate the highest values, at 1.55 and 1.56. Similar to the long-short returns, nonlinear (N-En) models outperform linear (L-En) models in terms of the annualized Sharpe ratio, with values of 1.54 and 1.17, respectively, further confirming the advantages of incorporating nonlinearity in the U.S. market.

Panels B and C of Table 4 present the portfolio performance results for the non-U.S. developed markets and the emerging markets, respectively. Consistent with our findings in the U.S. market, the ML models also deliver economically and statistically significant long-short performance in both non-U.S. markets.

In the non-U.S. developed market, the PCA model and the two-layer neural network (NN2) model stand out, generating the highest returns among the 10 ML models, with monthly returns of 0.97 % and 0.93 %, respectively. Additionally, the highest annualized Sharpe ratios in this market are achieved by the Partial Least Squares (PLS) model and the one-layer neural network (NN1) model, with Sharpe ratios of 1.49 and 1.60, respectively. These results highlight the robustness of certain ML models in capturing profitable trading signals in developed markets outside the U.S.

In the emerging markets, we find that the LASSO, PCA, and PLS models are equally effective, all generating the highest annualized Sharpe ratio of 1.14. Consistent with the results of the findings in Table 3, linear (L-En) models outperform nonlinear (N-En) models in the emerging markets. Specifically, the linear models demonstrate a stronger ability to produce higher average monthly returns and better risk-adjusted performance, as indicated by the annualized Sharpe ratios. This suggests that in emerging markets, the simplicity and structure of linear models may offer an advantage over more complex nonlinear approaches.

To mitigate concerns about the potential overweighting of small-capitalized and less liquid bonds, we additionally present the predictive performance results of value-weighted portfolios in Table 5, where the portfolio weights are determined by the amount issued. These results align with those of the equal-weighted portfolios shown in Table 4, as nearly all ML models produce both economically and statistically significant long-short portfolio profits. In the emerging markets context, linear models (L-En) continue to outperform nonlinear models (N-En), with monthly long-short portfolio returns of 1.30 % and 1.11 %, respectively, as demonstrated in the penultimate column of Panel C in Table 5. However, in the U.S. market, depicted in Panel A of Table 5, nonlinear models no longer generate higher monthly long-short portfolio returns than their linear counterparts. This outcome indicates that the incorporation of nonlinearity in ML models offers greater advantages for small-capitalized and less liquid bonds in the U.S. market when constructing equal-weighted portfolios, as evidenced by the results in Table 4.

3.3. Characteristic importance across markets

Next, we identify the bond characteristics that significantly enhance the predictive performance of ML models in each market group. Specifically, following the methodology of Gu et al. (2020), we first calculate the reduction in the time-series average monthly R -squared ($\bar{R}^2_{OOS}$) when setting all values of a given predictor to zero, while keeping the other predictors unchanged. This allows us to assess the impact of each characteristic on model performance. Variable importance is then defined as the relative reduction in $\bar{R}^2_{OOS}$. In this approach, the characteristic with the lowest R -squared reduction is assigned a variable importance of 0, while the characteristic with the highest reduction is assigned a variable importance of 1. To derive the overall rankings of characteristics across models, we rank the importance of each characteristic for each model and then average these rankings to produce a single rank. Figs. 2–4 present the overall rankings of characteristics across models for the U.S., non-U.S. developed, and emerging markets, respectively. Each

Table 4

Equal-weighted performance of ML portfolios. This table reports the out-of-sample performance of prediction-sorted portfolios in the U.S. market (Panel A), the non-U.S. developed markets (Panel B), and the emerging markets (Panel C) using 21 bond characteristics. At the end of each month, we sort bonds into deciles based on each model's one-month-ahead out-of-sample bond return predictions and construct an equal-weighted portfolio. We report the annualized Sharpe ratio for the long-short portfolio, as well as the time-series average monthly returns for each decile and the long-short portfolio. Newey-West t-statistics for long-short portfolio returns are reported in parentheses.

	Low	2	3	4	5	6	7	8	9	High	High-Low	High-Low SR
Panel A: U.S.												
LASSO	0.06	0.10	0.20	0.23	0.28	0.37	0.49	0.59	0.74	1.15	1.09(2.48)	1.13
RIDGE	0.12	0.16	0.19	0.24	0.29	0.37	0.48	0.56	0.68	1.11	0.99(2.26)	1.14
ENet	0.08	0.12	0.19	0.21	0.29	0.40	0.48	0.60	0.72	1.12	1.05(2.35)	1.09
PCA	0.08	0.12	0.18	0.22	0.30	0.35	0.47	0.58	0.65	1.25	1.17(2.52)	1.23
PLS	0.09	0.15	0.19	0.23	0.28	0.38	0.50	0.56	0.70	1.13	1.03(2.33)	1.17
RF	0.06	0.10	0.11	0.27	0.30	0.38	0.47	0.61	0.69	1.21	1.14(2.51)	1.22
GBRT	0.37	0.24	0.27	0.25	0.25	0.28	0.26	0.37	0.75	1.18	0.82(2.10)	1.45
NN1	0.08	0.17	0.19	0.25	0.31	0.38	0.46	0.50	0.70	1.17	1.09(2.61)	1.55
NN2	0.22	0.18	0.17	0.22	0.29	0.31	0.43	0.48	0.64	1.27	1.05(2.17)	1.56
NN3	0.14	0.13	0.17	0.24	0.33	0.35	0.44	0.56	0.67	1.18	1.04(2.21)	1.39
L-En	0.07	0.13	0.19	0.22	0.28	0.39	0.49	0.59	0.70	1.15	1.08(2.41)	1.17
N-En	0.09	0.15	0.20	0.22	0.30	0.36	0.43	0.51	0.70	1.23	1.14(2.55)	1.54
Panel B: Non-U.S. developed												
LASSO	0.13	0.14	0.16	0.23	0.23	0.31	0.34	0.46	0.51	0.95	0.83(1.88)	0.90
RIDGE	0.41	0.14	0.15	0.21	0.28	0.31	0.34	0.39	0.51	0.74	0.33(1.32)	0.48
ENet	0.13	0.12	0.16	0.23	0.24	0.29	0.39	0.42	0.46	1.03	0.90(1.77)	1.19
PCA	0.07	0.10	0.13	0.21	0.23	0.28	0.37	0.42	0.60	1.05	0.97(2.47)	0.86
PLS	0.04	0.12	0.17	0.21	0.26	0.33	0.33	0.53	0.55	0.94	0.90(1.87)	1.49
RF	0.11	0.12	0.13	0.17	0.21	0.33	0.34	0.45	0.58	1.03	0.92(2.47)	0.92
GBRT	0.37	0.31	0.29	0.31	0.25	0.30	0.31	0.36	0.39	0.58	0.22(1.72)	1.16
NN1	0.11	0.16	0.09	0.19	0.27	0.28	0.38	0.51	0.56	0.94	0.84(3.11)	1.60
NN2	0.00	0.13	0.19	0.24	0.28	0.35	0.38	0.41	0.56	0.94	0.93(3.02)	1.13
NN3	0.18	0.14	0.16	0.20	0.23	0.30	0.38	0.49	0.59	0.80	0.63(2.60)	1.00
L-En	0.06	0.10	0.12	0.25	0.24	0.33	0.39	0.45	0.53	0.99	0.93(1.99)	1.41
N-En	0.06	0.10	0.16	0.21	0.26	0.31	0.38	0.44	0.54	1.00	0.94(3.49)	1.44
Panel C: Emerging												
LASSO	0.13	0.15	0.18	0.24	0.27	0.32	0.44	0.46	0.51	1.33	1.20(3.19)	1.14
RIDGE	0.22	0.23	0.17	0.24	0.28	0.30	0.39	0.34	0.52	1.33	1.11(2.67)	1.12
ENet	0.16	0.17	0.18	0.23	0.28	0.32	0.42	0.44	0.51	1.32	1.16(2.96)	1.09
PCA	0.11	0.17	0.19	0.24	0.25	0.29	0.32	0.48	0.58	1.41	1.30(3.36)	1.14
PLS	0.15	0.21	0.19	0.23	0.25	0.31	0.41	0.42	0.51	1.36	1.20(3.09)	1.14
RF	0.20	0.17	0.22	0.19	0.32	0.33	0.40	0.38	0.49	1.33	1.13(3.18)	1.03
GBRT	0.32	0.15	0.17	0.17	0.26	0.34	0.41	0.45	0.45	1.30	0.99(2.95)	1.02
NN1	0.23	0.17	0.16	0.22	0.28	0.30	0.36	0.49	0.50	1.33	1.10(2.86)	1.05
NN2	0.23	0.14	0.18	0.18	0.29	0.32	0.42	0.43	0.51	1.33	1.10(3.07)	1.07
NN3	0.18	0.19	0.21	0.18	0.23	0.33	0.34	0.45	0.67	1.27	1.09(3.10)	0.99
L-En	0.14	0.19	0.16	0.25	0.27	0.30	0.42	0.42	0.51	1.37	1.23(3.16)	1.16
N-En	0.21	0.16	0.13	0.21	0.25	0.39	0.42	0.42	0.54	1.31	1.11(3.04)	1.02

column in the figures corresponds to a specific predictive model, with color gradients indicating the range of influence, from the most influential characteristics (dark) to the least influential (white).

In Fig. 2, bond momentum (*MOM12MBond*) is the most influential predictive determinant across nearly all ML models for the U.S. market, with the exception of three nonlinear models: RF, GBRT, and NN1. Additionally, another momentum measure, *MOM6MBond*, is highly instrumental in predicting bond returns, a finding that aligns with the work of Jostova et al. (2013). Long-term reversal (*LTR*) and downside risk (*Var5*) are the third and fourth most important predictors, respectively. Moreover, exposure measures to common factors, such as *BETA_BMkt* and *BETA_DEF*, are among the top 10 most significant characteristics in terms of predictive power.

Fig. 3 provides a detailed view of the overall rankings of characteristics in the non-U.S. developed markets. Here, we observe considerable disagreement among the ML models, particularly among the five nonlinear models, concerning the most important predictor. Nevertheless, downside risk (*Var5*) and volatility (*VOL*) emerge as the two most critical characteristics among 21 predictors we considered. Similar to the findings in the U.S. market, long-term reversal (*LTR*) is also identified as the third most important characteristic for predicting bond returns. Additionally, factor exposures such as *BETA_DEF* and *BETA_CRF*, along with momentum (*MOM12MBond*), rank among the top 10 most important predictors in this market.

Fig. 4 highlights the variable importance rankings in the emerging markets, where the results differ significantly from those in both the U.S. and non-U.S. developed markets. Illiquidity (*ILLIQ*) stands out as the most important predictive characteristic in this market. The other top predictors include momentum measures (*MOM12MBond* and *MOM6MBond*), downside risk (*Var5*), and duration (*DUR*). A particularly notable result is that four out of six factor exposures are ranked among the top 10 most important predictors in the emerging market. The critical role of illiquidity in predicting corporate bond returns can be attributed to the relatively low market-level liquidity in the emerging markets. This finding is consistent with the research of Chaieb et al. (2021a), who also identify a significantly higher liquidity premium in emerging equity markets compared to developed equity markets.

Table 5

Value-weighted performance of ML portfolios. This table reports the out-of-sample performance of prediction-sorted portfolios in the U.S. market (Panel A), the non-U.S. developed markets (Panel B), and the emerging markets (Panel C) using 21 bond characteristics. At the end of each month, we sort bonds into deciles based on each model's one-month-ahead out-of-sample bond return predictions and construct a value-weighted portfolio, using the amount issued as weights. We report the annualized Sharpe ratio for the long-short portfolio, as well as the time-series average monthly returns for each decile and the long-short portfolio. Newey-West t-statistics for the long-short portfolio returns are reported in parentheses.

	Low	2	3	4	5	6	7	8	9	High	High-Low	High-Low SR
Panel A: U.S.												
LASSO	0.05	0.11	0.20	0.24	0.32	0.43	0.56	0.67	0.81	1.38	1.33(2.29)	1.20
RIDGE	0.12	0.21	0.22	0.29	0.37	0.48	0.57	0.64	0.82	1.40	1.29(2.12)	1.22
ENet	0.08	0.14	0.21	0.22	0.32	0.46	0.55	0.67	0.83	1.39	1.31(2.24)	1.18
PCA	0.10	0.14	0.19	0.25	0.35	0.41	0.55	0.66	0.79	1.46	1.35(2.24)	1.21
PLS	0.10	0.20	0.22	0.28	0.35	0.47	0.56	0.66	0.83	1.40	1.30(2.20)	1.24
RF	0.05	0.11	0.12	0.28	0.32	0.39	0.49	0.62	0.70	1.45	1.41(3.01)	1.29
GBRT	0.28	0.25	0.27	0.25	0.23	0.26	0.24	0.38	0.79	1.42	1.13(2.19)	1.52
NN1	0.08	0.17	0.21	0.29	0.37	0.45	0.53	0.70	0.76	1.44	1.37(2.72)	1.44
NN2	0.21	0.19	0.20	0.25	0.34	0.35	0.51	0.57	0.77	1.43	1.22(2.22)	1.31
NN3	0.13	0.14	0.21	0.29	0.39	0.45	0.49	0.57	0.70	1.41	1.28(2.14)	1.24
L-En	0.08	0.17	0.22	0.27	0.33	0.47	0.56	0.68	0.82	1.44	1.35(2.23)	1.23
N-En	0.12	0.16	0.22	0.27	0.35	0.42	0.54	0.64	0.74	1.44	1.32(2.50)	1.30
Panel B: Non-U.S. developed												
LASSO	0.13	0.14	0.16	0.23	0.24	0.29	0.33	0.45	0.56	0.89	0.77(2.03)	0.84
RIDGE	0.35	0.09	0.14	0.19	0.28	0.30	0.34	0.41	0.54	0.73	0.38(1.59)	0.62
ENet	0.13	0.13	0.16	0.23	0.26	0.27	0.39	0.41	0.46	0.97	0.84(1.78)	1.09
PCA	0.07	0.10	0.13	0.22	0.24	0.27	0.39	0.44	0.65	1.08	1.01(2.30)	0.90
PLS	0.05	0.12	0.19	0.24	0.25	0.33	0.34	0.50	0.58	0.98	0.92(2.02)	1.35
RF	0.09	0.12	0.13	0.16	0.24	0.36	0.37	0.42	0.54	1.02	0.93(2.44)	0.96
GBRT	0.31	0.32	0.27	0.28	0.25	0.28	0.28	0.36	0.40	0.52	0.21(1.75)	1.12
NN1	0.19	0.16	0.11	0.19	0.28	0.30	0.39	0.49	0.59	0.91	0.72(3.65)	1.21
NN2	0.00	0.11	0.20	0.25	0.32	0.35	0.41	0.42	0.66	0.97	0.97(3.19)	1.15
NN3	0.18	0.15	0.16	0.21	0.24	0.34	0.44	0.52	0.62	0.81	0.63(2.76)	0.97
L-En	0.06	0.10	0.15	0.25	0.25	0.34	0.39	0.44	0.54	0.96	0.89(2.16)	1.27
N-En	0.09	0.10	0.17	0.23	0.28	0.32	0.38	0.43	0.63	0.97	0.89(3.51)	1.25
Panel C: Emerging												
LASSO	0.11	0.13	0.15	0.23	0.29	0.32	0.45	0.52	0.60	1.32	1.21(3.05)	1.00
RIDGE	0.23	0.21	0.17	0.25	0.30	0.33	0.40	0.37	0.55	1.42	1.19(2.73)	1.03
ENet	0.15	0.15	0.17	0.23	0.29	0.34	0.43	0.50	0.60	1.32	1.17(2.86)	0.96
PCA	0.11	0.17	0.18	0.24	0.27	0.31	0.33	0.50	0.69	1.46	1.34(3.42)	1.05
PLS	0.16	0.20	0.18	0.25	0.25	0.35	0.45	0.44	0.57	1.45	1.29(3.19)	1.05
RF	0.18	0.17	0.23	0.20	0.31	0.34	0.42	0.41	0.60	1.34	1.16(3.04)	0.95
GBRT	0.31	0.14	0.17	0.18	0.27	0.36	0.41	0.49	0.54	1.30	0.99(2.68)	0.88
NN1	0.24	0.19	0.19	0.24	0.30	0.31	0.37	0.51	0.50	1.35	1.11(2.87)	0.98
NN2	0.23	0.12	0.19	0.24	0.32	0.36	0.49	0.45	0.46	1.35	1.12(3.10)	0.99
NN3	0.17	0.19	0.20	0.21	0.24	0.39	0.39	0.50	0.71	1.25	1.08(3.05)	0.89
L-En	0.13	0.19	0.15	0.26	0.29	0.32	0.45	0.46	0.57	1.43	1.30(3.20)	1.07
N-En	0.18	0.17	0.13	0.24	0.29	0.40	0.45	0.48	0.57	1.29	1.11(2.97)	0.92

In conclusion, the characteristics that drive bond returns in the U.S. market differ substantially from those that impact bonds in non-U.S. markets. While downside risk and illiquidity are more influential in non-U.S. markets, momentum emerges as the most important predictor in the U.S. market. Despite these differences, there are also notable similarities in the return-characteristic relationships across the international corporate bond market. Specifically, downside risk, momentum, and exposure to bond risk factors—particularly default beta—are essential for predicting corporate bond returns in both U.S. and non-U.S. markets. This indicates that, while market-specific characteristics play a vital role, certain fundamental predictors retain their importance across different regions and market conditions.

3.4. Can the U.S. return-characteristics relationship be applied to other markets?

To further examine the differences in the characteristic-return relationships across these three markets, we use the methodology of Choi et al. (2025) to investigate whether models trained on U.S. data can outperform those utilizing characteristics specific to non-U.S. developed and emerging markets when predicting corporate bond returns. Specifically, we first estimate parameters and hyperparameters using the training and validation samples from the U.S. market. We then apply these U.S.-derived parameters and hyperparameters to predict corporate bond returns in the non-U.S. developed and emerging markets, using market-specific characteristics from these two regions as predictors.

Panel A of Table 6 presents the results for the non-U.S. developed markets. The first and second rows report the out-of-sample R^2 (R^2_{Oos}) and the differences in R^2_{Oos} when using U.S. parameters, respectively. The findings reveal that seven out of 10 market-specific models experience a reduction in R^2_{Oos} compared to models based on U.S. parameters. This suggests that, in many cases,

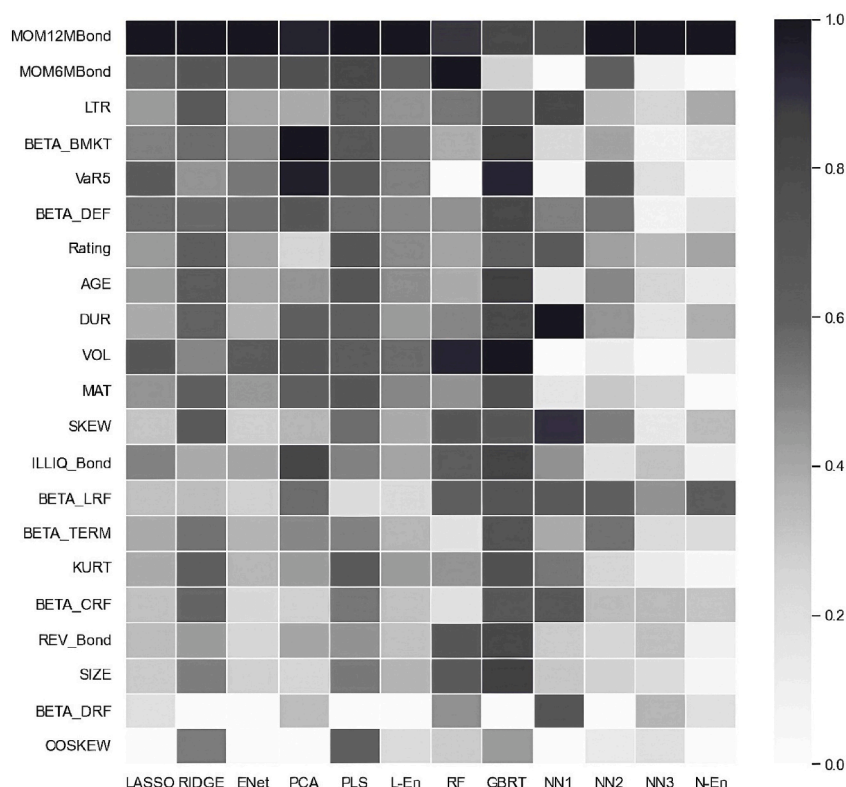


Fig. 2. Characteristic importance in the U.S. market. This figure presents the overall rankings of 21 bond characteristics based on their variable importance in each model in the U.S. market. To measure variable importance, we first calculate the reduction of time-series average monthly R -squared $\bar{R}_{OOS,t}^2$ from setting all values of a given predictor to zero, while keeping the remaining predictors unchanged. Variable importance is defined as the relative reduction of $\bar{R}_{OOS,t}^2$. In this way, the characteristic with the lowest R -squared reduction has a variable importance of 0, while the characteristic with the highest R -squared reduction has a variable importance of 1. To get the overall rankings of characteristics across models, we rank the importance of each characteristic for each model and then average them into a single rank. Columns correspond to individual models, and color gradients within each column indicate the most influential (dark) to least influential (white) characteristics.

market-specific models outperform U.S.-based models in the non-U.S. developed market. Notably, the last two columns of Panel A indicate that the improvements in R_{OOS}^2 when using U.S. parameters are primarily driven by nonlinear models (N-En), with a positive 0.22 % of R_{OOS}^2 difference. This highlights the potential advantage of these nonlinear models in effectively leveraging U.S. market information. Panel B of Table 6 reports the results for the emerging markets. In contrast to the non-U.S. developed markets, all models using U.S. parameters underperform their market-specific counterparts, with R_{OOS}^2 differences ranging from -3.48 % for LASSO to -0.63 % for GBRT. The findings suggest that U.S.-based models are less effective in capturing the unique dynamics of emerging markets.

Overall, the results in Table 6 demonstrate that the emerging markets diverge more significantly from the U.S. market in terms of the relationship between returns and characteristics compared to the non-U.S. developed markets. This highlights the importance of tailoring models to the specific characteristics of emerging markets, as U.S.-based models may not fully capture the nuances of these regions.

3.5. Integration of international bond markets

In this subsection, we expand our analysis to investigate the integration of corporate bond markets. While substantial research has focused on equity market integration (e.g., Bekaert and Harvey, 1995; Carrieri et al., 2007; Bekaert et al., 2009; Pukthuanthong and Roll, 2009; Bekaert et al., 2011; Carrieri et al., 2013; Akbari et al., 2020), the integration of international corporate bond markets remains relatively underexplored.⁸

Theoretically, perfectly integrated markets imply that assets with identical risk profiles should have the same expected returns, regardless of the markets in which they are traded. This concept, grounded in the law of one price, necessitates the absence of cross-

⁸ For a literature review of equity market integration, refer to Akbari and Ng (2020).

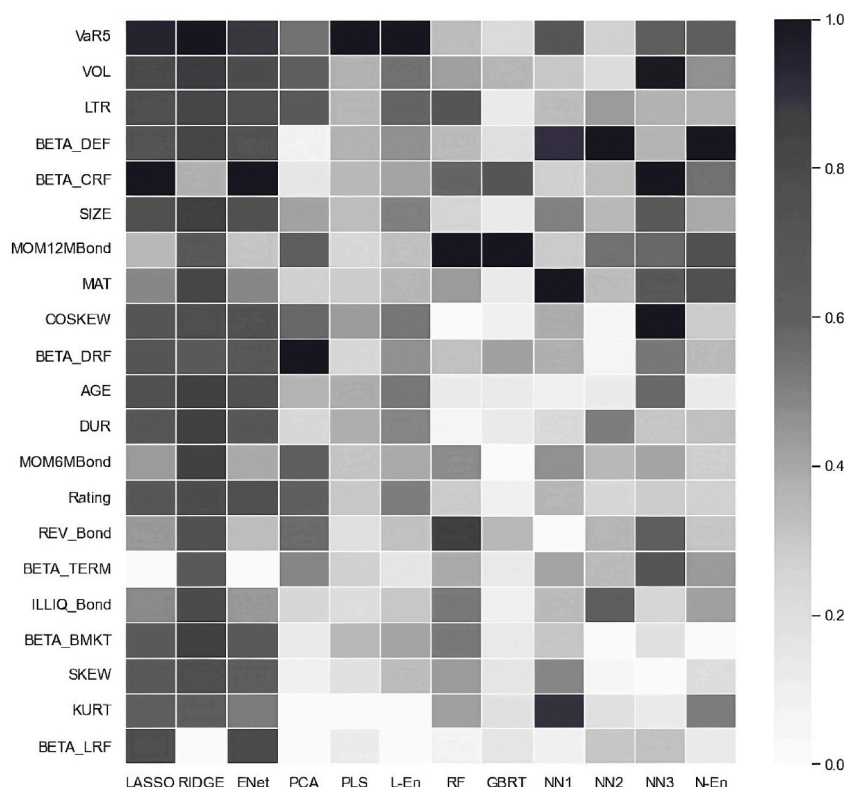


Fig. 3. Characteristic importance in the non-U.S. developed markets. This figure presents the overall rankings of 21 bond characteristics based on their variable importance in each model in the non-U.S. developed markets. To measure variable importance, we first calculate the reduction of time-series average monthly R -squared $\bar{R}_{OOS,t}^2$ from setting all values of a given predictor to zero, while keeping the remaining predictors unchanged. Variable importance is defined as the relative reduction of $\bar{R}_{OOS,t}^2$. In this way, the characteristic with the lowest R -squared reduction has a variable importance of 0, while the characteristic with the highest R -squared reduction has a variable importance of 1. To get the overall rankings of characteristics across models, we rank the importance of each characteristic for each model and then average them into a single rank. Columns correspond to individual models, and color gradients within each column indicate the most influential (dark) to least influential (white) characteristics.

market arbitrage opportunities (Chen and Knez, 1995). Market integration is advantageous as it reduces the cost of capital, broadens investment opportunities, and fosters economic growth. However, both explicit barriers (such as legal restrictions on ownership and institutional constraints) and implicit barriers (such as asymmetric information and regulatory differences) can impact cross-market risk-sharing and thus affect the degree of market integration. There is a wealth of literature investigating cross-economy variations in market integration (e.g., Bekaert and Harvey, 1995; Carrieri et al., 2007; Bekaert et al., 2011; Carrieri et al., 2013; Chaieb et al., 2020).

In international finance, it is commonly assumed that local asset returns are influenced by both local and global factors. As financial markets become more integrated, an asset's returns are expected to be more closely tied to global factors rather than local ones (Chari and Henry, 2004; Pukthuanthong and Roll, 2009). Based on this premise, our measure of market integration is derived from the change in R_{OOS}^2 after incorporating global factor information. Specifically, we include bonds' exposures to five U.S. bond factors (*BMKT*, *DEF*, *LRF*, *CRF*, and *DRF*) as new characteristics to test whether and to what extent these predictors improve our predictions of corporate returns in non-U.S. developed and emerging markets.⁹ The U.S. market is chosen to represent global markets for two primary reasons: (1) the U.S. bond market constitutes 81 % of our observations, and (2) given the U.S.'s prominent role in global economies and financial markets, numerous studies have identified U.S. market conditions as significant sources of risk for global financial markets (e.g., Rapach et al., 2013; Miranda-Agrippino and Rey, 2020).¹⁰

Panel A of Table 7 presents the results for the non-U.S. developed markets. The findings reveal that eight out of our 10 ML models (excluding GBRT and NN2) show improved predictive performance, as indicated by higher R_{OOS}^2 , after incorporating the five U.S. factor

⁹ Due to a lack of information on the returns of the long-term government bonds in non-U.S. markets, we have already applied U.S. TERM factor to construct *BETA_TERM* in both the non-U.S. developed and emerging markets.

¹⁰ We obtain similar results when using "true" global factors, following Fama and French (2017) and Hollstein (2022), instead of U.S.-based global proxies. Results are available upon request.

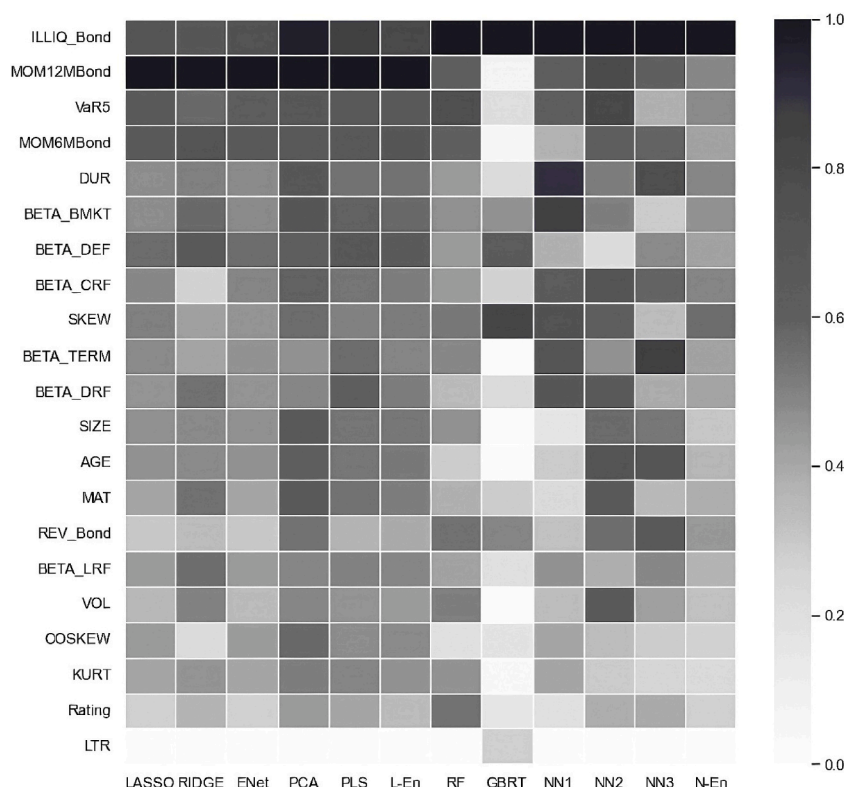


Fig. 4. Characteristic importance in the emerging markets. This figure presents the overall rankings of 21 bond characteristics based on their variable importance in each model in the emerging markets. To measure variable importance, we first calculate the reduction of time-series average monthly R -squared $\bar{R}_{OOS,t}^2$ from setting all values of a given predictor to zero, while keeping the remaining predictors unchanged. Variable importance is defined as the relative reduction of $\bar{R}_{OOS,t}^2$. In this way, the characteristic with the lowest R -squared reduction has a variable importance of 0, while the characteristic with the highest R -squared reduction has a variable importance of 1. To get the overall rankings of characteristics across models, we rank the importance of each characteristic for each model and then average them into a single rank. Columns correspond to individual models, and color gradients within each column indicate the most influential (dark) to least influential (white) characteristics.

Table 6

Predictive performance of ML models using U.S. parameters. This table reports the out-of-sample predictive performance of ML models in the non-U.S. developed markets (Panel A) and the emerging markets (Panel B), using parameters derived from the training and validation samples in the U.S. market. We use 21 bond characteristics to predict corporate bond returns. For each market group, we report R_{OOS}^2 and the change in R_{OOS}^2 when applying U.S. parameters. The ML models include LASSO, ridge regression (RIDGE), elastic net (ENet), principal component analysis (PCA), partial least squares (PLS), random forest (RF), gradient boosted regression trees (GBRT), and neural networks with one to three layers (NN1–NN3). The L-En (N-En) model averages the forecasts of five linear (nonlinear) models, including LASSO, RIDGE, ENet, PCA, and PLS (RF, GBRT, NN1, NN2, and NN3).

	LASSO	RIDGE	ENet	PCA	PLS	RF	GBRT	NN1	NN2	NN3	L-En	N-En
Panel A: Non-U.S. developed												
R_{OOS}^2 with U.S. parameters	2.53	2.65	2.54	2.88	3.33	4.21	4.03	1.30	2.07	3.12	2.87	3.75
R_{OOS}^2 Difference	-1.47	0.03	-2.04	-1.28	-1.44	1.24	1.66	-2.26	-1.72	-0.11	-1.51	0.22
Panel B: Emerging												
R_{OOS}^2 with U.S. parameters	1.52	1.94	1.77	1.57	2.13	2.09	2.38	2.67	1.94	2.02	1.89	3.02
R_{OOS}^2 Difference	-3.48	-3.20	-3.20	-3.45	-3.15	-2.66	-0.63	-2.11	-2.97	-2.65	-3.28	-1.79

exposures. Notably, the improvement is more significant for linear models (L-En) compared to nonlinear models (N-En), with an R_{OOS}^2 difference of 0.36 % versus 0.28 %. Panel B reports the results for the emerging markets, where we find that seven out of 10 models (excluding GBRT, NN3, and NN2) exhibit increased R_{OOS}^2 when U.S. factor exposures are included. Similar to the non-U.S. developed markets, linear models show a larger increase in R_{OOS}^2 for emerging markets. However, the average R_{OOS}^2 change across all 10 models is lower for emerging markets compared to non-U.S. developed markets (0.31 % vs. 0.16 %). Overall, the results from Table 7 indicate that incorporating U.S. market information can marginally enhance predictive performance for non-U.S. markets, with a more pronounced effect for non-U.S. developed markets.

Table 7

Bond market integration: by market group. This table reports the out-of-sample predictive performance of ML models in the non-U.S. developed markets (Panel A) and the emerging markets (Panel B), incorporating bonds' exposures to U.S. factors as additional characteristics. The U.S. factors include *BMKT*, *DEF*, *LRF*, *CRF*, and *DRF*, which are constructed in this paper. We include 21 bond characteristics and the newly added five U.S. factor exposures to predict corporate bond returns. For each market group, we report R^2_{OOS} and the change in R^2_{OOS} after adding the five U.S. factor exposures. The ML models include LASSO, ridge regression (RIDGE), elastic net (ENet), principal component analysis (PCA), partial least squares (PLS), random forest (RF), gradient boosted regression trees (GBRT), and neural networks with one to three layers (NN1–NN3). The L-En (N-En) model averages the forecasts of five linear (nonlinear) models, including LASSO, RIDGE, ENet, PCA, and PLS (RF, GBRT, NN1, NN2, and NN3).

	LASSO	RIDGE	ENet	PCA	PLS	RF	GBRT	NN1	NN2	NN3	L-En	N-En
Panel A: Non-U.S. developed												
R^2_{OOS} with U.S. Factor	4.03	3.48	4.79	4.42	5.17	3.32	2.37	3.88	3.57	4.12	4.74	3.81
R^2_{OOS} Difference	0.03	0.86	0.21	0.26	0.40	0.35	0.00	0.31	−0.22	0.89	0.36	0.28
Panel B: Emerging												
R^2_{OOS} with U.S. Factor	5.33	5.26	5.28	5.32	5.53	5.61	2.73	5.02	4.37	4.66	5.48	4.92
R^2_{OOS} Difference	0.34	0.12	0.31	0.30	0.25	0.86	−0.28	0.23	−0.54	0.00	0.31	0.11

Considering that explicit and implicit barriers contribute to the cross-economy differences in the degree of market integration, we further calculate the percentage change in R^2_{OOS} after including U.S. factor exposures for each economy. This percentage change serves as a measure of the economy-level integration index.¹¹ Table 8 presents the integration levels for 12 developed economies and 10 emerging markets. Consistent with the findings in Table 7, the average level of integration is higher among developed markets compared to emerging markets. This pattern mirrors the observed higher integration in equity markets for developed markets (e.g., Akbari et al., 2020). Among developed markets, Canada exhibits the highest degree of integration with the global or U.S. bond market (118.39 %), while Singapore shows the lowest (−2.76 %). Japan, as the world's third-largest economy, is notably segmented from the international bond market, with the second-lowest integration (−0.92 %). Among emerging markets, Mexico demonstrates the highest degree of integration with the global or U.S. bond market (49.67 %), whereas Russia has the lowest (−52.80 %). Despite being the second-largest economy globally, China has the fourth-lowest integration (−0.96 %). The high integration levels for Canada and Mexico could be attributed to their geographic proximity to the U.S., where U.S. investors are more influential and more likely to invest (Brogaard et al., 2020).

Prior studies identify several determinants of market integration. Akbari et al. (2021) highlight financial development, the quality of the information environment and financial openness as key drivers of cross-country variation in equity market integration. Similarly, Chaieb et al. (2020) find that sovereign bond market integration is positively associated with political stability, credit quality, low inflation and inflation risk, and market liquidity. Our finding that non-U.S. developed markets tend to be more integrated than emerging markets is consistent with these earlier results. Emerging markets typically face lower levels of financial development, weaker information environments, limited openness, and higher inflation risk and illiquidity—factors that can impede full integration with global markets.

It is important to note, however, that our five-year out-of-sample window limits our ability to empirically examine the role of country-level macroeconomic fundamentals in shaping integration dynamics. A more thorough investigation would require longer return histories and richer cross-country economic data.

Bonds are predominantly held by institutional investors. The corporate bond market is less liquid than the equity market, which may result in different dynamics in bond integration compared to equities. Fig. 5 illustrates the relationship between bond integration, as analyzed in this study, and equity integration, as reported by Akbari et al. (2020). The figure reveals a very weak correlation between bond integration and equity integration, highlighting the distinct nature of bond market integration compared to equity markets.

3.6. Bond-stock integration

So far, our analysis has exclusively utilized bond characteristics to predict corporate bond returns. However, integrating equity characteristics could potentially enhance our predictive models, given that both stocks and bonds are claims on the same underlying assets of a firm. Merton's (1974) structural credit risk model suggests that bonds and stocks should be jointly priced, meaning they should exhibit a degree of integration if financial frictions do not limit arbitrage opportunities. To investigate bond–stock integration, we follow Bali et al. (2022) and expand our models by incorporating 24 additional stock characteristics to assess their impact on predictive performance.

Panel A of Table 9 presents the results for the U.S. market, where the inclusion of stock characteristics leads to a general decrease in predictive performance, as evidenced by lower R^2_{OOS} values across all models. This outcome aligns with findings from Bali et al. (2022), which suggest that bonds and stocks in the U.S. market are not strongly integrated. Conversely, Panels B and C show somewhat different results for non-U.S. developed and emerging markets. In these regions, some models, such as NN3 in non-U.S. developed

¹¹ To reduce the noise of R^2_{OOS} , we require at least 50 bonds for each economy. We average the forecasts of 10 machine learning models to estimate R^2_{OOS} to construct a unified integration measure.

Table 8

Bond market integration: by economy. This table reports bond market integration by economy. Bond market integration for each economy is measured by the relative percentage change in R^2_{OOS} after adding exposures to U.S. bond factors (including $BMKT$, DEF , LRF , CRF , and DRF) as new predictors. We train our ML models and make predictions for each economy separately, requiring at least 50 bonds per economy to ensure adequate observations for calculating R^2_{OOS} . For brevity, we report the R^2_{OOS} based on the average predictions across 10 ML models.

Developed Markets	Num of Bonds	Integration (%)	Emerging Markets	Num of Bonds	Integration (%)
Canada	590	118.39	Mexico	166	49.67
Ireland	57	14.37	India	141	34.62
Italy	80	8.08	Indonesia	55	6.36
United Kingdom	563	4.35	Chile	57	1.61
Australia	452	3.75	South Korea	139	1.45
Hong Kong	125	2.75	Brazil	234	-0.45
Germany	159	0.98	China	700	-0.96
Switzerland	58	0.08	Turkey	69	-4.22
Netherlands	70	-0.35	United Arab Emirates	101	-16.34
France	164	-0.49	Russia	97	-52.80
Japan	221	-0.92			
Singapore	54	-2.76			
DEV Mean	216	12.35	EMG Mean	176	1.89

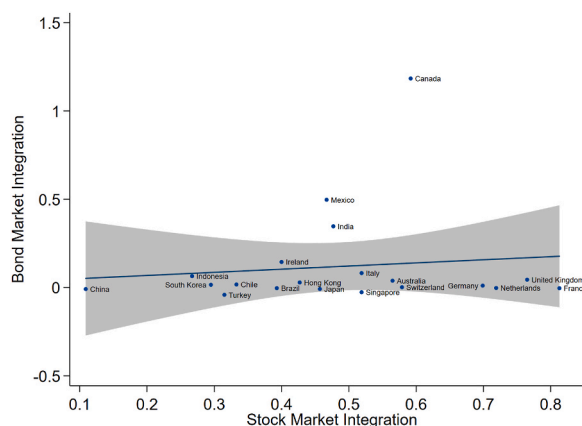


Fig. 5. Bond market integration and stock market integration. This figure presents the correlation between bond market integration and stock markets integration. Bond market integration for each economy is measured by the relative change in R^2_{OOS} after adding the exposures to U.S. bond factors (including $BMkt$, DEF , LRF , CRF , and DRF) as new predictors. We average the forecasts of 10 ML models to estimate R^2_{OOS} . Stock market integration data is sourced from Akbari et al. (2020), who decompose stock returns into cash-flow revisions and risk-pricing revisions. The measure of stock market cross-economy integration reflects the extent to which an economy's risk-pricing revisions comove with global risk-pricing revisions.

Table 9

Bond-stock integration: by market group. This table reports the out-of-sample predictive performance of ML models in the U.S. market (Panel A), the non-U.S. developed markets (Panel B), and the emerging markets (Panel C) after adding stock characteristics. We use 21 bond characteristics and 24 stock characteristics to predict corporate bond returns. For each market group, we report R^2_{OOS} and the change in R^2_{OOS} after including the 24 stock characteristics. The ML models include LASSO, ridge regression (RIDGE), elastic net (ENet), principal component analysis (PCA), partial least squares (PLS), random forest (RF), gradient boosted regression trees (GBRT), and neural networks with one to three layers (NN1–NN3). The L-En (N-En) model averages the forecasts of five linear (nonlinear) models, including LASSO, RIDGE, ENet, PCA, and PLS (RF, GBRT, NN1, NN2, and NN3).

	LASSO	RIDGE	ENet	PCA	PLS	RF	GBRT	NN1	NN2	NN3	L-En	N-En
Panel A: U.S.												
R^2_{OOS} with Stock Chas	3.42	3.33	3.46	3.83	3.66	3.68	3.12	4.07	3.69	4.28	3.69	4.40
R^2_{OOS} Difference	-0.20	-0.34	-0.21	-0.33	-0.16	-0.08	-0.41	-0.10	-0.69	-0.32	-0.16	-0.10
Panel B: Non-U.S. developed												
R^2_{OOS} with Stock Chas	3.68	0.72	4.19	3.93	4.06	2.82	2.17	2.89	3.63	3.49	4.00	3.44
R^2_{OOS} Difference	-0.33	-1.91	-0.39	-0.23	-0.71	-0.15	-0.20	-0.68	-0.16	0.25	-0.38	-0.08
Panel C: Emerging												
R^2_{OOS} with Stock Chas	4.17	4.35	4.37	4.43	4.49	4.50	2.57	4.88	4.77	4.48	4.64	4.76
R^2_{OOS} Difference	-0.82	-0.80	-0.60	-0.59	-0.79	-0.25	-0.45	0.10	-0.14	-0.19	-0.53	-0.05

markets and NN1 in emerging markets, demonstrate an improvement in R_{OOS}^2 when stock characteristics are included. This suggests that there may be a degree of bond-stock integration in these markets, which is not as pronounced in the U.S. market.

These results highlight potential cross-economy differences in bond-stock integration. To quantify this, we measure bond-stock integration for each economy by evaluating the change in R_{OOS}^2 after incorporating stock characteristics. The results in Table 10 demonstrate that bond-stock integration is generally higher in developed markets compared to emerging markets. Among developed markets, Canada exhibits the highest degree of bond-stock integration (114.45 %), while Ireland shows the lowest (−42.05 %). In emerging markets, Mexico has the lowest degree of bond-stock integration (−166.66 %), whereas Turkey has the highest (30.11 %).

These findings underscore that the integration of bonds and stocks varies considerably between developed and emerging markets, as well as across different economies. The extent of this integration is potentially shaped by various factors, including the development of financial markets and the degree of market friction.

Our additional (unreported) results indicate that stock-bond integration increased in all three markets following the onset of the COVID-19 pandemic. Several economic forces may explain this shift. First, the pandemic triggered a globally synchronized economic shock and policy response, likely strengthening the co-movement between equity and bond markets by amplifying the influence of shared macroeconomic fundamentals. Second, during periods of heightened uncertainty, equity markets tend to react more quickly to new information than bond markets, making stock characteristics more useful for predicting bond returns. Third, the growing use of cross-asset investment strategies and algorithmic trading may have further deepened linkages between the two markets. Finally, aggressive central bank interventions in response to the pandemic may have reduced the informativeness of bond-specific characteristics (e.g., credit spreads), thereby increasing the relative importance of stock-based information. Consistent with this interpretation, we observe a sharp decline in predictive performance after the onset of COVID-19 (see Table 11).

4. Extended analyses and robustness tests

4.1. Time-varying return predictability

We next investigate the time-varying predictive performance of ML models and explore the underlying economic mechanisms driving this variation. As an initial step, we examine the impact of the COVID-19 pandemic, a major macroeconomic shock that profoundly influenced global financial markets. To assess the time variation in return predictability, we compare the out-of-sample performance (R_{OOS}^2) of our models before and after the onset of COVID-19. Our final sample covers approximately 10 years of data from March 2012 to December 2021. To ensure equal sample lengths, we define the pre-COVID period as spanning from May 2018 to February 2020, and the post-COVID period from March 2020 to December 2021. For brevity, we report the R_{OOS}^2 based on the average predictions across 10 ML models. As shown in Table 11, we observe a significant decline in predictive performance post-COVID across U.S., non-U.S. developed, and emerging markets. This decline is more pronounced for investment-grade bonds. For example, in the U.S., the average R_{OOS}^2 drops from 7.89 % pre-COVID to 3.53 % post-COVID—a decrease of 4.36 percentage points. The reduction is 5.80 % for investment-grade bonds versus 2.79 % for non-investment-grade bonds. Several factors may explain this decline. First, extraordinary central bank interventions (e.g., the Federal Reserve's SMCCF and PMCCF) may have distorted market risk pricing and reduced the informativeness of traditional bond characteristics. Second, heightened uncertainty and shifts in investor behavior may have weakened the link between fundamentals and future returns. Finally, structural changes in firm risk profiles during the recovery could have also played a role.

In addition to COVID-19, broader market conditions may also influence ML model performance. Previous studies document that return predictability tends to be stronger under adverse market conditions. For example, Dangi and Halling (2012) report enhanced predictability for the S&P 500 during economic downturns. Avramov et al. (2023) find that predictive performance improves amid

Table 10

Bond-stock integration: by economy. This table reports bond-stock integration by economy. Bond-stock integration for each economy is measured by the relative percentage change in R_{OOS}^2 after adding stock characteristics as new predictors. We train our ML models and make predictions for each economy separately, requiring at least 50 bonds per economy to ensure adequate observations for calculating R_{OOS}^2 . For brevity, we report the R_{OOS}^2 based on the average predictions across 10 ML models.

Developed Markets	Num of Bonds	Integration (%)	Emerging Markets	Num of Bonds	Integration (%)
Canada	590	114.45	Turkey	69	30.11
Italy	80	50.28	South Korea	139	17.46
France	164	21.69	Indonesia	55	9.69
Japan	221	3.70	Chile	57	−0.87
Germany	159	2.59	Brazil	234	−12.38
Netherlands	70	0.63	United Arab Emirates	101	−23.59
Australia	452	−0.03	China	700	−67.46
Singapore	54	−2.83	Russia	97	−70.94
Hong Kong	125	−9.11	India	141	−119.50
Switzerland	58	−9.22	Mexico	166	−166.66
United Kingdom	563	−34.47			
Ireland	57	−42.05			
DEV Mean	216	7.97	EMG Mean	176	−40.42

Table 11

Predictive performance before and after the COVID-19 pandemic. This table reports the out-of-sample predictive performance (R^2_{OOS}) of ML models across different markets before and after the COVID-19 pandemic, covering the U.S., non-U.S. developed markets, and emerging markets. We also present the absolute difference in R^2_{OOS} between these two periods. To ensure comparability, the pre- and post-COVID periods are of equal length, with the pre-COVID sample spanning 2018m5 to 2020m2 and the post-COVID sample covering 2020m3 to 2021m12. For each market group, bonds are further sorted into investment-grade (IG) bonds—those with ratings of 10 or below (i.e., BBB– or higher)—and non-investment-grade (NIG) bonds, which have ratings of 11 or above (i.e., BB + or lower). Bolded values indicate which market state yields a higher R^2_{OOS} . We use 21 bond characteristics to predict corporate bond returns across the three market groups. For brevity, we report the R^2_{OOS} based on the average predictions across 10 ML models.

	U.S.			Non-U.S. developed			Emerging		
	Before	After	Diff	Before	After	Diff	Before	After	Diff
All Bonds	7.89	3.53	4.36	9.68	3.23	6.45	7.02	4.23	2.79
IG	8.16	2.36	5.80	11.40	3.11	8.29	11.32	2.73	8.59
NIG	4.70	1.91	2.79	3.00	1.86	1.14	5.35	3.98	1.37

heightened market volatility and reduced liquidity. In the Treasury bond market, [Bianchi et al. \(2021\)](#) emphasize the influence of investor risk aversion on forecast accuracy. Similarly, [Feng et al. \(2025\)](#) show that corporate bond return predictability in the U.S. increases in market conditions characterized by weak economic fundamentals, low liquidity, elevated sentiment, and high-risk aversion.

Motivated by these findings, we examine four market condition variables: the business conditions index of [Aruoba et al. \(2009\)](#) (BC), the market liquidity measure of [Pástor and Stambaugh \(2003\)](#) (LIQ), the investor sentiment index of [Huang et al. \(2015\)](#) (SNT), and the risk aversion index of [Bekaert et al. \(2022\)](#) (RA). Due to the notable drop in R^2_{OOS} following the COVID-19 outbreak, we focus on the pre-COVID sample period (up to 2020m2). For each macroeconomic variable, we classify observations into high and low states using the median as a threshold and calculate the corresponding R^2_{OOS} for each group. As shown in [Table 12](#), we observe that predictive performance is significantly stronger across all three markets (U.S., non-U.S. developed, and emerging) and for both investment-grade and non-investment-grade bonds when business conditions and liquidity are low, and investor sentiment is high. An exception arises in the non-U.S. developed market, where ML models perform better under high business conditions for non-investment-grade bonds. Overall, under these three favorable predictive conditions, R^2_{OOS} increases by 1.74 %–14.88 % across all bonds and markets.

For risk aversion, we observe that ML model performance is better during high-risk-aversion periods in the U.S. and non-U.S. developed markets, with increases in R^2_{OOS} of 5.47 % and 8.42 %, respectively. In emerging markets, although the overall R^2_{OOS} is slightly higher during low-risk-aversion periods (+1.06 %), this is driven primarily by non-investment-grade bonds. For investment-grade bonds, predictive performance remains stronger during periods of high-risk aversion, with R^2_{OOS} increasing by 10.14 %.

Among the four macroeconomic variables, investor sentiment is the most influential factor in the two developed markets, with

Table 12

Predictive performance under different market conditions. This table reports the out-of-sample predictive performance (R^2_{OOS}) of ML models across different markets prior to the COVID period (i.e., before 2020m3, covering the U.S., non-U.S. developed markets, and emerging markets). The results are shown under different market conditions, defined as high (above the median) and low (below the median) states. We also report the absolute difference in R^2_{OOS} between these two states. Panel A includes all bonds, Panel B focuses on investment-grade bonds (with ratings of 10 or below, i.e., BBB– or higher), and Panel C covers non-investment-grade bonds (with ratings of 11 or above, i.e., BB + or lower). Four market condition variables are considered: the business conditions index from [Aruoba et al. \(2009\)](#) (BC), the market liquidity measure from [Pástor and Stambaugh \(2003\)](#) (LIQ), the investor sentiment index from [Huang et al. \(2015\)](#) (SNT), and the risk aversion index from [Bekaert et al. \(2022\)](#) (RA). Bolded values indicate which market state yields a higher R^2_{OOS} . We use 21 bond characteristics to predict corporate bond returns across the three market groups. For brevity, we report the R^2_{OOS} based on the average predictions across 10 ML models.

	U.S.			Non-U.S. developed			Emerging		
	Low	High	Diff	Low	High	Diff	Low	High	Diff
Panel A: All bonds									
BC	7.64	5.71	1.93	8.63	6.89	1.74	12.45	4.14	8.31
LIQ	9.23	3.87	5.36	9.45	5.26	4.19	14.78	−0.10	14.88
SNT	1.40	9.77	8.37	0.67	12.27	11.60	4.63	11.14	6.51
RA	2.80	8.27	5.47	1.95	10.37	8.42	8.80	7.74	1.06
Panel B: Investment-grade									
BC	9.21	5.28	3.93	11.31	7.34	3.97	12.58	5.92	6.66
LIQ	11.12	3.04	8.08	11.24	6.18	5.06	9.72	7.59	2.13
SNT	1.52	11.42	9.90	1.02	13.53	12.51	3.88	11.87	7.99
RA	2.93	8.65	5.72	0.68	12.35	11.67	2.14	12.28	10.14
Panel C: Non-investment-grade									
BC	4.93	4.28	0.65	3.96	5.59	1.63	12.76	2.39	10.37
LIQ	5.86	3.25	2.61	5.06	4.21	0.85	16.51	−3.17	19.68
SNT	0.65	6.15	5.50	1.92	7.30	5.38	5.15	9.95	4.80
RA	3.46	5.37	1.91	6.37	3.62	2.75	9.65	5.54	4.11

changes in R_{OOS}^2 of 8.37 % in the U.S. market and 11.60 % in non-U.S. developed markets, followed by risk aversion, which leads to R_{OOS}^2 increases of 5.47 % and 8.42 %, respectively. In contrast, in emerging markets, liquidity plays the most prominent role, with a 14.88 % improvement in R_{OOS}^2 between high- and low-liquidity periods, followed by business conditions, which contribute an 8.31 % increase.

Overall, these findings confirm that bond return predictability varies over time and is significantly influenced by economic conditions, market liquidity, investor sentiment, and risk aversion. The results are consistent with prior evidence from U.S. equity, Treasury bond, and corporate bond markets (Dangl and Halling, 2012; Bianchi et al., 2021; Avramov et al., 2023; Feng et al., 2025), and offer novel insights from a global corporate bond perspective.

4.2. Portfolio performance net of transaction costs

We further examine whether the substantial profits generated by our ML-based long-short strategies remain economically meaningful after accounting for transaction costs. Prior research highlights the non-negligible role of transaction costs in corporate bond trading. For instance, Edwards et al. (2007) document that trading costs vary significantly with trade and issue size, with estimates ranging between 20 and 80 basis points. Similarly, Bali et al. (2022) emphasize the importance of trading frictions in bond markets and propose the use of effective spreads as a practical proxy for these costs. Unfortunately, our international dataset does not contain the necessary transaction-level details to implement these estimation methods directly.

To address this challenge, and in line with the approach of Feng et al. (2025), we evaluate the robustness of our strategy returns under assumed transaction cost levels from 0 to 80 bps. Specifically, we report both turnover ratios and transaction-cost-adjusted performance metrics, including monthly average returns and annualized Sharpe ratios. For brevity, we focus on the portfolios formed using the L-En and N-En models.

As reported in Table 13, the turnover ratios remain moderate across all three markets (U.S., non-U.S. developed, and emerging), with values below 70 % for both the L-En and N-En models. Importantly, even after applying the highest transaction cost assumption (80 bps), the long-short portfolios remain profitable. For example, using the L-En model, the monthly average return holds at 0.77 % in both the U.S. and emerging markets, with corresponding annualized Sharpe ratios of 0.83 and 0.73, respectively. In the non-U.S. developed markets, the N-En model delivers a 0.43 % monthly return and a 0.67 Sharpe ratio under the same cost assumption.

Overall, these findings suggest that our ML-based strategies are economically viable after accounting for realistic transaction costs.

4.3. Robustness test: removing economies with limited coverage

Although our original corporate bond database spans from 2010 to 2021, our final sample begins in 2012. This is because we require at least 24 monthly return observations within each 36-month rolling window to conduct meaningful return predictions. Not every economy but most economies in our sample cover 2012 to 2021. Specifically, 91 % (20 out of 22) of the non-U.S. developed economies have data coverage beginning in 2012, and all of these developed economies have data through 2021. Among emerging markets, 56 % (35 out of 63) have data starting in 2012. Importantly, those emerging economies with later starting dates tend to have a small number of observations (ranging from 1 to 543). Moreover, 78 % (49 out of 63) of emerging economies cover at least half of the total sample period (2012–2021).

In our return predictability analysis by market, we aggregate bonds from all emerging economies into a single group rather than analyzing each country individually. Therefore, including economies that do not cover the entire 10-year period is akin to including bonds with varying histories within the same market, which we believe is reasonable. For analyses where individual country-level inference is required (e.g., bond market or bond-stock market integration by economy in Tables 8 and 10), we require each economy to have at least 50 bonds to ensure robustness.

To further address concerns that differences in sample periods across economies may affect our main findings on cross-market predictive performance, we conduct a robustness check by excluding economies that do not span the full sample period. In Table 14, we report the out-of-sample R_{OOS}^2 of the L-En and N-En models under three scenarios: (1) no restriction on sample coverage, (2) economies with at least 8 years of data, and (3) economies with full 10-year coverage. The results show that imposing stricter coverage requirements leads to only a slight reduction in predictive performance in both non-U.S. developed and emerging markets, with decreases in R_{OOS}^2 ranging from 0 % to 0.4 %. Importantly, our main conclusions remain unchanged: linear models tend to perform better in non-U.S. markets, and overall, ML models achieve higher predictive performance in non-U.S. markets compared to the U.S. market.

4.4. Robustness test: predictive performance under varying training and validation periods

We initially adopted a 3-year training period and a 2-year validation period (3T2V), following the general practice in the literature (e.g., Gu et al., 2020; Leipold et al., 2022), which typically allocates approximately half of the sample for model training and validation. To address concerns that our findings might be sensitive to this specific choice, we conduct robustness checks using alternative training-validation combinations, including 4T2V, 5T2V, 6T2V, 4T3V, and 5T3V. To ensure comparability, we report the out-of-sample R^2 (R_{OOS}^2) for the L-En and N-En models during the COVID period (2020m2–2021m12) in Table 15. The results show that the L-En model is insensitive to the length of the training period when the validation window is held constant. For the N-En model, R_{OOS}^2 varies by only 0.04 %–0.48 % across different training periods and markets. Changes in the validation window affect both models, but

Table 13

Impact of transaction costs on long-short portfolio performance. This table provides the results of an examination of the impact of transaction costs on the out-of-sample performance of long-short portfolios constructed using predictions from ML models (L-En and N-En) across three markets. We report the portfolio turnover, as well as the monthly average returns and annualized Sharpe ratios after accounting for transaction costs ranging from 0 to 80 basis points. We use 21 bond characteristics to predict corporate bond returns across these three markets. The L-En (N-En) model averages the forecasts of five linear (nonlinear) models, including LASSO, RIDGE, ENet, PCA, and PLS (RF, GBRT, NN1, NN2, and NN3).

Turnover (%)		Monthly Average Returns (%)					Annualized Sharpe Ratio				
		0bp	20bp	40bp	60bp	80bp	0bp	20bp	40bp	60bp	80bp
Panel A: U.S.											
L-En	39.41	1.08	1.01	0.93	0.85	0.77	1.17	1.08	1.00	0.91	0.83
N-En	54.75	1.14	1.03	0.92	0.81	0.70	1.54	1.39	1.25	1.10	0.95
Panel B: Non-U.S. developed											
L-En	68.16	0.93	0.79	0.65	0.52	0.38	1.41	1.20	0.99	0.78	0.58
N-En	63.08	0.94	0.81	0.68	0.56	0.43	1.44	1.25	1.06	0.86	0.67
Panel C: Emerging											
L-En	57.26	1.23	1.12	1.00	0.89	0.77	1.16	1.05	0.95	0.84	0.73
N-En	55.87	1.11	1.00	0.88	0.77	0.66	1.02	0.91	0.81	0.71	0.61

Table 14

Predictive performance with varying sample coverage. This table reports the out-of-sample predictive performance (R^2_{OOS}) of ML models—L-En and N-En—in non-U.S. developed markets (Panel A) and emerging markets (Panel B). The results are presented under different sample coverage requirements for each economy: no filtering, at least 8 years of data, and full coverage of the 10-year sample period. We use 21 bond characteristics to predict corporate bond returns across these two markets. The L-En (N-En) model averages the forecasts of five linear (nonlinear) models, including LASSO, RIDGE, ENet, PCA, and PLS (RF, GBRT, NN1, NN2, and NN3).

	No Filtering	At least 8 years	10 years
Panel A: Non-U.S. developed			
L-En	4.38	4.38	4.35
N-En	3.52	3.52	3.51
Panel B: Emerging			
L-En	5.17	4.98	4.80
N-En	4.81	4.57	4.41

the resulting variation in R^2_{OOS} remains below 0.5 %. These findings suggest that our main results are robust to different training and validation period specifications.

Table 11 shows that predictive performance declines significantly following the onset of COVID-19. While we have discussed several potential explanations for this structural break, the robustness check results in Table 15 suggest that this decline is not driven by the choice of training or validation periods. As more recent data become available, retraining the models on a post-COVID sample could improve predictive performance by better reflecting the evolving relationships between bond characteristics and returns.

4.5. Discussion of USD-denominated vs. local currency bonds

The availability of local-currency bonds in emerging markets is severely limited when using the IHS Markit data, primarily due to the lack of credit rating information in Datastream. After applying the same filtering criteria as in our main analysis, only 2950 observations remain for local currency bonds in emerging markets, compared to 81,282 for USD-denominated bonds. Given this limited sample size and the lack of consistent credit rating coverage, our preceding analysis focuses on USD-denominated bonds. Nonetheless, it is important to discuss potential differences in predictive performance and market integration between USD-denominated and local currency bonds.

Theoretically, we expect ML models to perform better when predicting USD-denominated bond returns for several reasons. First, USD-denominated bonds are typically issued by larger, more internationally active firms that exhibit higher disclosure standards and more consistent financial reporting, enhancing data quality and model training. Second, USD bonds tend to be more liquid and actively traded, resulting in more accurate and timely pricing. In contrast, local currency bonds—especially in emerging markets—are often illiquid or priced using indicative quotes, introducing noise that can hinder predictability. Third, local currency bonds are exposed to currency risk, and the resulting volatility may mask the underlying credit risk and return patterns that ML models are designed to detect. By focusing on USD-denominated bonds, we reduce the confounding effects of exchange rate fluctuations. Fourth, the global USD corporate bond market is more structured and integrated, with more stable pricing dynamics across countries and over time. This improves the transferability of learned return patterns and enhances out-of-sample model performance. Finally, credit ratings and other essential bond characteristics are more consistently available for USD-denominated bonds, particularly in non-U.S. markets where data coverage for local currency issues is sparse or incomplete.

Supporting this view, unreported results indicate that ML models deliver stronger predictive performance for USD-denominated bonds relative to local currency bonds in non-U.S. developed markets. For similar reasons, we also expect lower levels of market

Table 15

Predictive performance under varying training and validation periods. This table reports the out-of-sample predictive performance (R^2_{OOS}) of ML models—L-En and N-En—across different markets during the COVID period (2020m3–2021m12): the U.S. (Panel A), non-U.S. developed markets (Panel B), and emerging markets (Panel C). Results are presented under various combinations of training and validation periods: 3 years training and 2 years validation (3T2V, used in the preceding analysis), 4T2V, 5T2V, 6T2V, 4T3V, and 5T3V. We use 21 bond characteristics to predict corporate bond returns across these three markets. The L-En (N-En) model averages the forecasts of five linear (nonlinear) models, including LASSO, RIDGE, ENet, PCA, and PLS (RF, GBRT, NN1, NN2, and NN3).

	3T2V	4T2V	5T2V	6T2V	4T3V	5T3V
Panel A: U.S.						
L-En	3.08	3.08	3.08	3.08	2.86	2.86
N-En	3.87	3.69	4.06	3.75	3.76	3.80
Panel B: Non-U.S. developed						
L-En	3.96	3.96	3.96	3.96	4.39	4.39
N-En	2.34	2.10	2.17	1.86	2.27	1.96
Panel C: Emerging						
L-En	4.44	4.44	4.44	4.44	3.98	3.98
N-En	3.89	3.84	3.56	3.74	3.31	3.31

integration for local currency bonds, as integration is largely driven by cross-border investor participation, which is more common in USD-denominated markets.

5. Conclusion

There is a growing number of studies devoted to the cross-section of corporate bond returns in the U.S. market. However, research in the international corporate bond market has been limited due to data availability. Utilizing a novel international dataset, this paper is, to our knowledge, the first to introduce ML techniques to thoroughly analyze the out-of-sample predictability of international corporate bond returns.

Dividing the international corporate bond market into three categories—U.S., non-U.S. developed, and emerging markets—we find that ML models make an important difference: they generate positive out-of-sample R -squared and sizable long-short portfolio strategy profits across all three markets. Interestingly, the most important return predictors of corporate bonds in non-U.S. markets are significantly different from those in the U.S. market, with downside risk and illiquidity being much more influential in non-U.S. markets.

Our results also reveal a significant decline in predictive accuracy following the onset of COVID-19 across U.S., non-U.S. developed, and emerging markets, suggesting a structural shift in the informativeness of traditional bond characteristics. Furthermore, we find that bond return predictability is higher during periods marked by deteriorating business conditions, reduced market liquidity, elevated investor sentiment, and heightened risk aversion. Among the four macroeconomic variables, investor sentiment emerges as the most influential factor in the two developed markets, whereas market liquidity plays the most prominent role in emerging markets.

We also investigate bond cross-market integration and bond-stock integration. Our findings suggest that non-U.S. developed markets are more integrated with the global or U.S. bond markets compared to emerging markets. Moreover, there are wide cross-economy variations in the degree of bond integration, which are only weakly related to equity integration. Although corporate bonds are not well-integrated with stocks in the U.S., we find that additional stock characteristics help improve the predictive performance of ML models in some economies. Theoretically, bond cross-market integration and bond-stock integration both rely on the ability to freely invest across economies and bond-stock markets within an economy. Thus, any economy-level frictions (explicit or implicit barriers) that hinder capital flows and limit arbitrage can potentially lead to cross-economy differences in bond cross-market integration and bond-stock integration. For future research, it would be valuable to develop structural models to combine with ML and AI techniques to further explain cross-economy differences in bond return predictability in global markets.

CRedit authorship contribution statement

Delong Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lei Lu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhen Qi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Guofu Zhou:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Appendix. Characteristics descriptions

A. Corporate bond characteristics (21)

1. **Credit Rating (*Rating*)**. We obtain bond-level rating information from Bloomberg. Credit ratings are turned to conventional linear numeric scores where a higher numerical score means higher credit risk (AAA = 1, AA+ = 2, ..., C = 21). When two ratings are provided by Moody's and Fitch, the ratings are averaged.
2. **Time-to-maturity (*MAT*)**. The number of years to maturity.
3. **Age (*Age*)**. The number of years since bonds' issuance.
4. **Duration (*DUR*)**. The Macaulay duration, obtained from IHS Markit data.
5. **Issuance size (*Size*)**. The natural logarithm of bond amount issued.
6. **Illiquidity (*ILLIQ_Bond*)**. Following Amihud and Mendelson (1986), we first calculate relative bid-ask spread as ask price minus bid price, divided by the average of bid and ask prices for each day. Then, we take the mean of daily values in a month to get a monthly illiquidity proxy for each bond.
7. **Volatility (*VOL*)**. We estimate historical volatility using monthly returns for the past 36 months, requiring a minimum of 24 observations.
8. **Skewness (*SKEW*)**. We estimate historical skewness using monthly returns for the past 36 months, requiring a minimum of 24 observations.
9. **Kurtosis (*KURT*)**. We estimate historical Kurtosis using monthly returns for the past 36 months, requiring a minimum of 24 observations.
10. **Downside risk (*Var5*)**. We measure the downside risk of corporate bonds using the second lowest daily return observation over the past 36 trading months, with a minimum requirement of 24 observations. For convenience of interpretation, we multiply the original measure by -1 .
11. **Short-term reversal (*REV_Bond*)**. We measure short-term reversal using bond returns in the previous month.
12. **Six-month momentum (*MOM6Bond*)**. Lagged two-month to lagged six-month cumulative returns.
13. **Twelve-month momentum (*MOM12Bond*)**. Lagged two-month to lagged twelve-month cumulative returns.
14. **Long-term reversal (*LTR*)**. Following Bali et al. (2021), it is defined as lagged 13-month to lagged 48-month cumulative returns.
15. **Bond market beta (*BETA_BMKT*)**. Following Bali et al. (2022), we estimate the bond market beta of individual bonds on bond market excess return using the rolling sample of the past 36 months, with a minimum requirement of 24 observations. Bond market excess return is computed as the value-weighted (amount issuance as weight) average returns of all corporate bonds for each of the three markets minus the risk-free rate.
16. **Term beta (*BETA_TERM*)**. Following Fama and French (1993) and Crawford et al. (2019), we estimate term beta of individual bonds on *TERM* factor using the rolling sample of the past 36 months (requiring a minimum of 24 observations), while controlling for the bond market factor. *TERM* factor is defined as $TERM = r_{SBTSY10} - r_f$, where $r_{SBTSY10}$ is the return on long-term government bonds based on *SBTSY10* index (a 10-year constant-maturity price index) from Bloomberg.
17. **Default beta (*BETA_DEF*)**. Following Fama and French (1993) and Crawford et al. (2019), we estimate the default beta of individual bonds on *DEF* factor using the rolling sample of the past 36 months (requiring a minimum of 24 observations), while controlling for the bond market factor. *DEF* factor is calculated as $DEF = r_{Sample} - r_{SBTSY10}$, where r_{Sample} is the return on investment-grade corporate bonds with more than 10 years to maturity for each of the three markets, weighted by amount issued.
18. **Downside risk beta (*BETA_DRF*)**. We estimate the downside risk beta of individual bonds on downside risk factor using the rolling sample of the past 36 months (requiring a minimum of 24 observations), while controlling for the bond market factor. Downside risk factor is the value-weighted average return difference between the highest *Var5* portfolio (quintile 5) and the lowest *Var5* portfolio (quintile 1).¹²
19. **Credit risk beta (*BETA_CRF*)**. We estimate the credit risk beta of individual bonds on credit risk factor using the rolling sample of the past 36 months (requiring a minimum of 24 observations), while controlling for the bond market factor. Credit risk factor is the value-weighted average return difference between the highest rating portfolio (quintile 5) and the lowest rating portfolio (quintile 1).
20. **Illiquidity risk beta (*BETA_LRF*)**. We estimate illiquidity risk beta of individual bonds on illiquidity risk factor using the rolling sample of the past 36 months (requiring a minimum of 24 observations), while controlling for the bond market factor. Illiquidity risk factor is the value-weighted average return difference between the highest illiquidity portfolio (quintile 5) and the lowest illiquidity portfolio (quintile 1).
21. **Co-skewness (*COSKEW*)**. Following Boyer et al. (2010), co-skewness is estimated based on the following rolling regressions of the past 36 months:

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \gamma_i R_{m,t}^2 + \varepsilon_{i,t},$$

¹² We do not use independent sorts based on credit rating because there are a relatively small number of bonds in non-U.S. markets.

where $R_{i,t}$ is monthly excess return of bond i , $R_{m,t}$ is monthly bond market excess return for each of the three markets, and γ_i is co-skewness (systematic skewness) of bond i .

B. Stock characteristics (24)

Following [Feng et al. \(2025\)](#), we first construct 18 stock characteristics based on the variable definitions from [Hou et al. \(2020\)](#).¹³ We also select the six most important stock characteristics from [Fig. 5](#) of [Gu et al. \(2020\)](#).

B.1. Stock Characteristics from [Hou et al. \(2020\)](#)

1. **Standard unexpected earnings (SUE)**. The change in split-adjusted quarterly earnings per share from its value one year ago scaled by standard deviation of this change over the prior two years.
2. **Cumulative abnormal returns around earnings announcement dates (ABR)**. The cumulative abnormal daily stock return around the latest quarterly earnings announcement date.
3. **Revisions in analyst earnings forecasts (RE)**. The six-month moving average of past changes in analysts' forecasts.
4. **Twelve-month momentum (MOM12Stock)**. Lagged two-month to lagged 12-month cumulative monthly returns.
5. **Book-to-market (BM)**. Yearly book equity to monthly market equity.
6. **Earnings-to-price (EP)**. Quarterly earnings to monthly price.
7. **Cashflow-to-price (CFP)**. Quarterly cashflow to monthly price.
8. **Sales-to-price (SP)**. Quarterly sales to monthly price.
9. **Asset growth rate (AGR)**. Quarterly asset growth rate.
10. **Quarterly equity issuance (NI)**. Quarterly equity issuance.
11. **Accruals (ACC)**. It is measured as net income minus total operating, investing, and financing cash flow plus sales of stocks minus stock repurchases and dividends.
12. **Return on asset (ROA)**. Quarterly return on quarterly asset.
13. **Seasonality (Ra1M)**. Returns in month $t - 12$.
14. **Seasonality (Rn1M)**. Average returns from $t - 11$ to $t - 1$.
15. **Market equity (ME)**. Monthly market equity.
16. **Stock variance (SVAR)**. We estimate historical volatility using daily returns for the past month.
17. **CAPM beta (BETA)**. Following [Fama and French \(1993\)](#), we estimate the market beta of individual stocks on value-weighted average market excess returns using the rolling sample of the past 36 months.
18. **Short-term reversal (REV_Stock)**. We measure short-term reversal using the stock return in the previous month.

B.2. Stock characteristics from [Gu et al. \(2020\)](#)

1. **Six-month momentum (MOM6Stock)**. Lagged two-month to lagged six-month cumulative monthly returns.
2. **Idiosyncratic risk (IDIOVOL)**. Standard deviation of residuals of monthly returns on monthly value-weighted market excess returns for the past 36 months.
3. **Maximum daily return (MAXRET)**. Maximum daily returns in a month.
4. **Amihud illiquidity (ILLIQ_Stock)**. Monthly average of daily absolute return divided by dollar volume.
5. **Bid-ask spread (BASPREAD)**. Monthly average of daily bid-ask spread divided by average of daily bid and ask price.
6. **Share turnover (TURN)**. Average monthly trading volume for the most recent three months scaled by shares outstanding.

Data availability

Data will be made available on request.

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¹³ Because we study the international markets, we obtain stock pricing and accounting information from Datastream rather than Compustat, as in [Hou et al. \(2020\)](#). We change the constructions slightly for some stock characteristics if needed.

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